

Aerial Robotics

FIELD AND SERVICE ROBOTICS

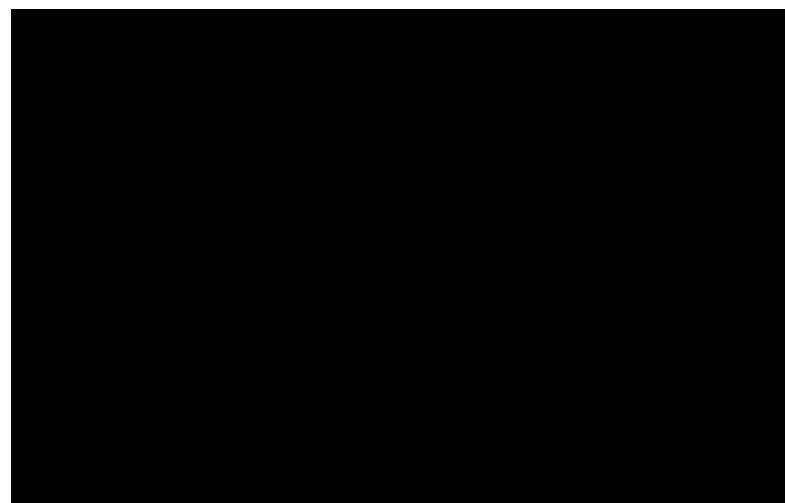
 **DIE** **UNIVERSITA'** **DEGLI STUDI DI**
TI. **NA** **POLI FEDERICO II**
DIPARTIMENTO DI INGEGNERIA ELETTRICA
E TECNOLOGIE DELL'INFORMAZIONE

www.prisma.unina.it

- Aerial robotics has become a new frontier in field and service robotics

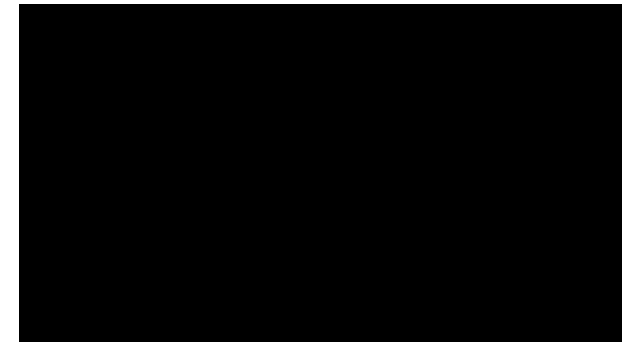
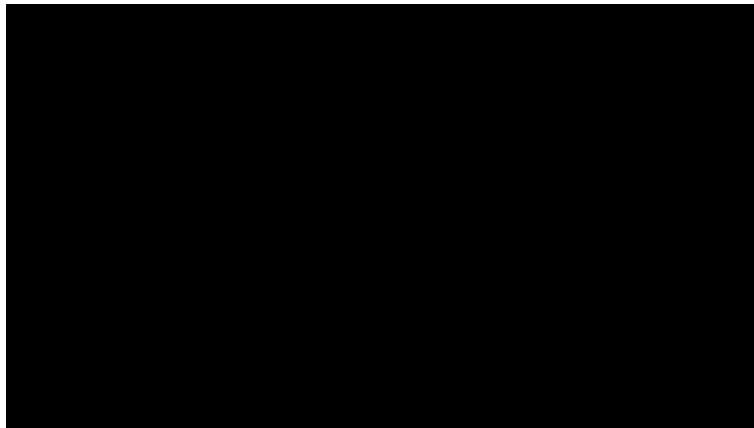


- Really useful?



- Business opportunity

Delivering?



Industry?



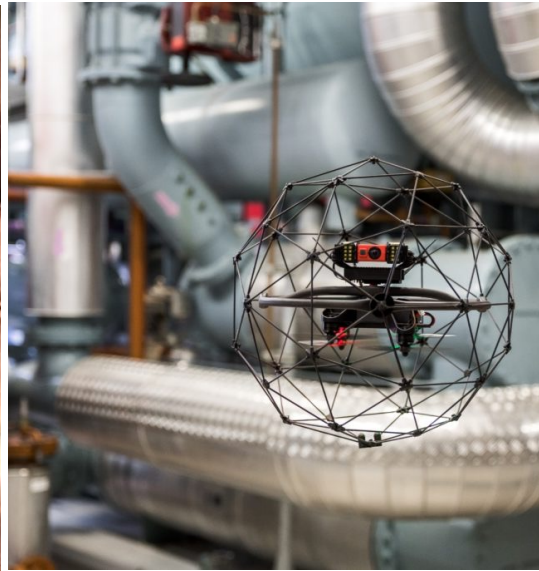
Marketing?



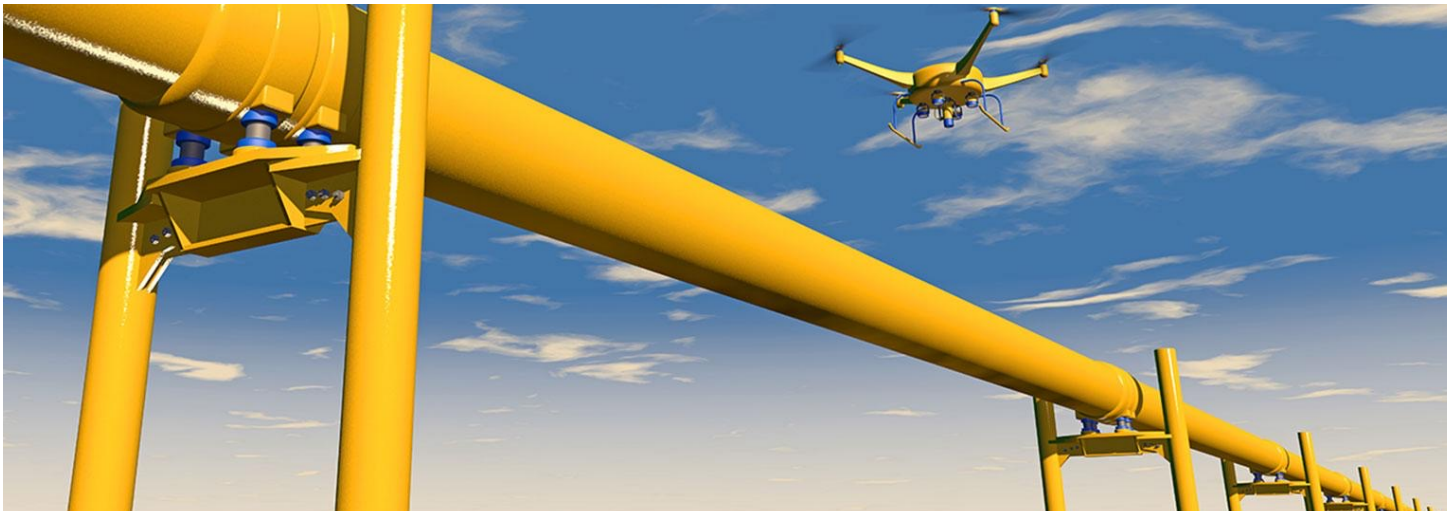
- Aerial robotics applications
 - Transportation
 - Manipulation
 - Inspection and maintenance (I&M)
 - Repairing
 - Monitoring
 - Patrolling
 - Exploration
 - Intervention in devastated areas
 - ...

- Oil and gas facilities
 - What is possible:
 - 3D mapping of existing plants
 - Visual inspection
 - Thermographic inspection
 - Direct measurements (e.g., thickness)
 - First intervention and monitoring
 - What is needed:
 - ATEX compliant solutions
 - Power remote-handling for maintenance and (dis)assembling

- Drone for inspection in confined spaces



- Drone for long pipe inspection



- Drone for thickness measurements



■ History

- Leonard Da Vinci's notes
- First autonomous flight of an **unmanned aerial vehicle** (UAV) on 1896
 - Number 5 vehicle, designed by Samuel P. Langley, that travelled 400 meters
 - Number 6 vehicle travelled 1600 meters
- In 1916, during the First World War, the Aerial Target vehicle was radio controlled (RC)
- Hewitt-Sperry vehicle in 1916



- Before the Second World War the airplanes were converted into autonomous vehicles controlled by autopilots
 - The first society, the *Radioplane Company*, specialized in building was founded by Reginald Denny
 - 1500 radio-controlled helicopters were built for the Second World War
- Technological development, smaller processors, lightweight sensor measurement systems, global navigations systems and so on boosted UAVs growth
 - In Japan, Yamaha Motor company developed an unmanned helicopter for irrigation in 1983
 - An autonomous UAV flying with GPS was out in 1988
 - It observed volcanic activities at Usu-zan in 2000

■ Acronyms

- UAV – Unmanned Aerial Vehicle
- UAS – Unmanned Aerial System
- RPV – Remotely Piloted System
- ROA – Remotely Operated Aircraft
- UVS – Unmanned Vehicle System

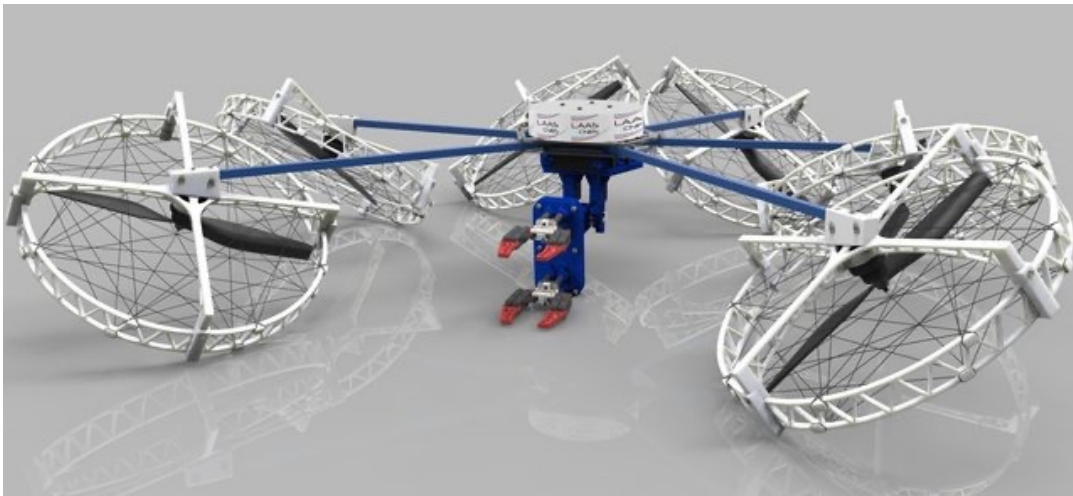
- Fixed-wing UAVs
 - They require a track to take off and land
 - High robustness, high cruise velocity



- Rotary-wing UAVs
 - Vertical take off and landing (VTOL) vehicles
 - Hovering
 - High manoeuvrability
 - Helicopters, quadrotor, hexarotors,



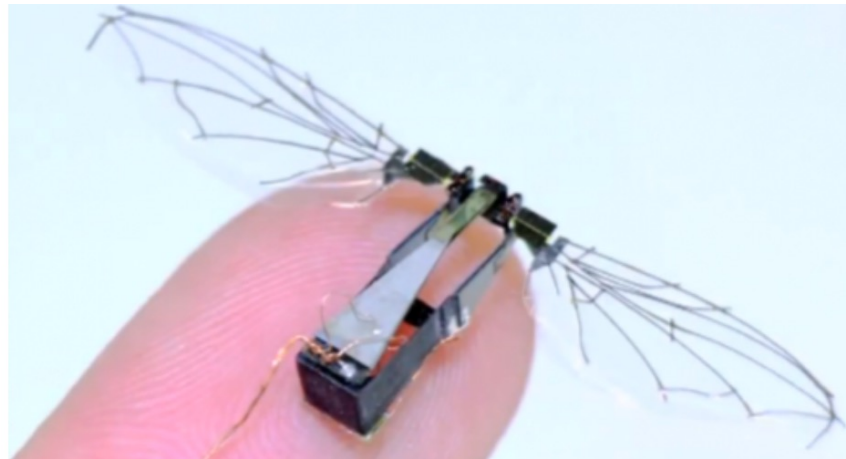
- Tilted (passive) or tilting (active) configurations



- Blimp
 - Lighter-than-air
 - High robustness
 - Low velocity



- Flapping wings
 - Bio-inspired



■ Taxonomy about dimension and flight time

- HALE (High Altitude Long Endurance)
- MALE (Medium Altitude Long Endurance)
- Tactical UAVs
- Small and man-portable UAVs



- MAVs (Micro Aerial Vehicles)



- Performance of aerial robots depends on dimensions and flying mechanisms (wings, rotors, ...)
- Effects of dimensions on a quadrotor
 - Blade velocity $v \propto \sqrt{R}$
 - Lift force $F \propto R^3$
 - Mass $m \propto R^3$
 - Inertia $I \propto R^5$
 - Angular acceleration $\dot{\omega} \propto \frac{1}{R}$



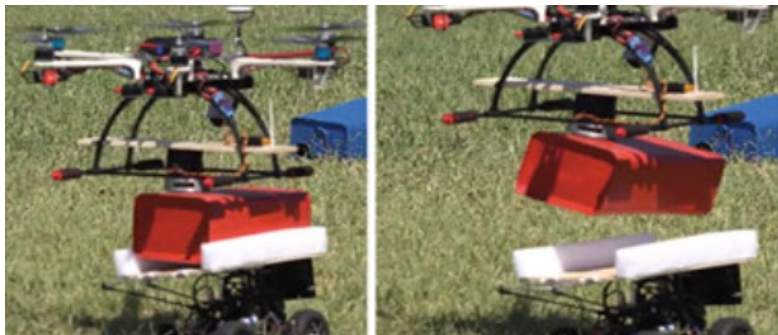
- Propulsion systems
 - Jet
 - Internal combustion
 - Spacecraft propulsion
 - Electrical
- Flight regimes
 - Hovering
 - Tailsitters robot
 - Cruising flight



- Physical interaction while flying is relevant for many applications
 - Interaction with other flying objects
 - Refuelling
 - Object transportation
 - Capture of drones for security reasons
 - Docking
 - Maintenance and repairing



- Interaction with ground objects
 - Mobile objects
 - Picking mobile targets
 - Landing on mobile platforms
 - Fixed objects
 - Picking fixed targets
 - Cleaning
 - Contact inspection
 - Manipulation

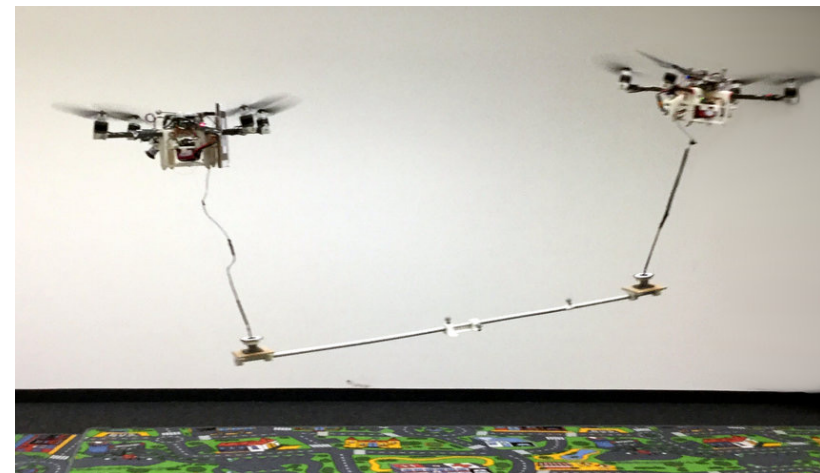




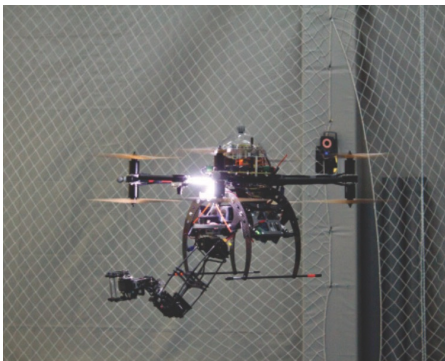
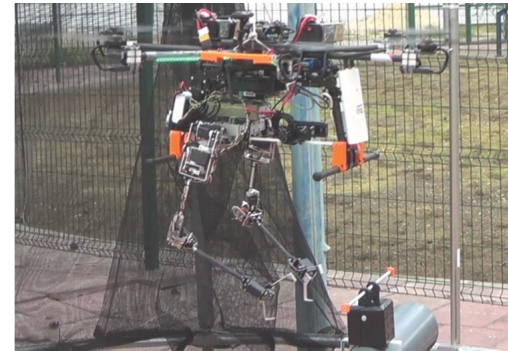
■ Problems

- Stability of the aircraft during the interaction
- Accuracy
 - Trajectory tracking
 - Positioning with respect to targets
 - Grasping
 - Physical contact with objects
- Aerodynamic perturbations due to proximity to surfaces
- Payload required to carry devices for physical interaction
- Required flight time
- Reactivity to cancel the effect of wind perturbations flying close to objects
- Planning

- Aerial manipulation is a special case of physically interacting UAVs with the environment
- Aerial manipulation can be achieved through
 - Flying hands
 - Object or rigid tool directly attached to the UAV
 - The object is linked or tethered



- Unmanned aerial manipulators (UAMs)



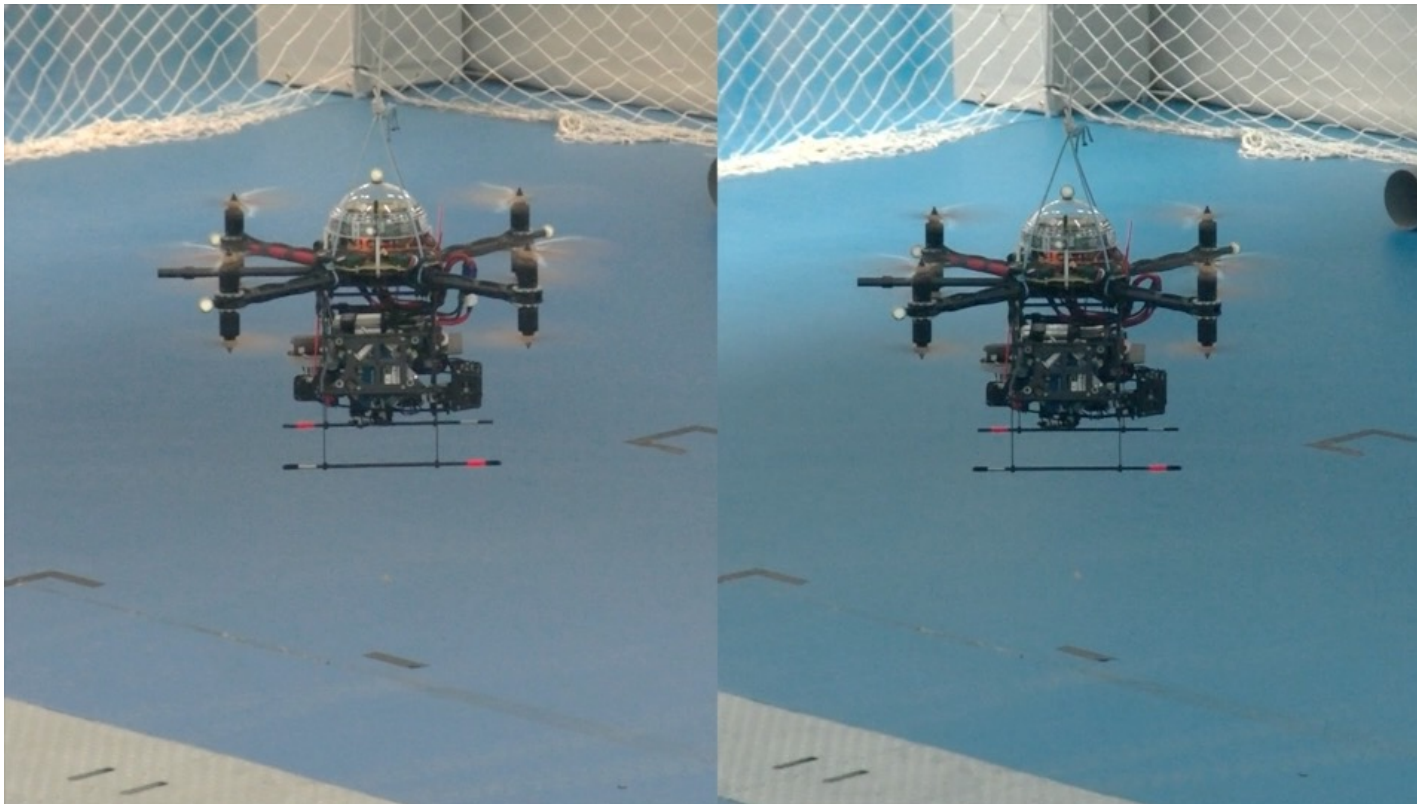


- UAMs are constituted by the following elements
 - Floating base (UAV)
 - Robotic arm
 - Gripper or multi-fingered hand (end-effector)
 - Sensors (cameras, laser scanners,...)
- UAMs can be classified in several ways
 - Number of DoFs of the arm, or number of arms on the device (single arm, dual-arm,...)
 - Mechanism of the joints
 - Control modality of the motors
 - Resulting configuration
- The mounted robot arm creates issues due to the coupling dynamics
- Two ways to control the UAM
 - Centralized approach
 - The UAV and the robotic arm are seen as a unique entity
 - Decentralized approach
 - The UAV and the robotic arm are seen as independent systems



Reactiveness of the proposed estimator
Interaction with a human operator

www.youtube.com/watch?v=iHKtHF0LF-w



www.youtube.com/watch?v=4I_2Pa_xOUM



www.youtube.com/watch?v=5hR-xYMbK50



Development of a Control Framework to Autonomously Install Clip Bird Diverters on High-Voltage Lines

Simone D'Angelo, Francesca Pagano, Fabio Ruggiero, Vincenzo Lippiello

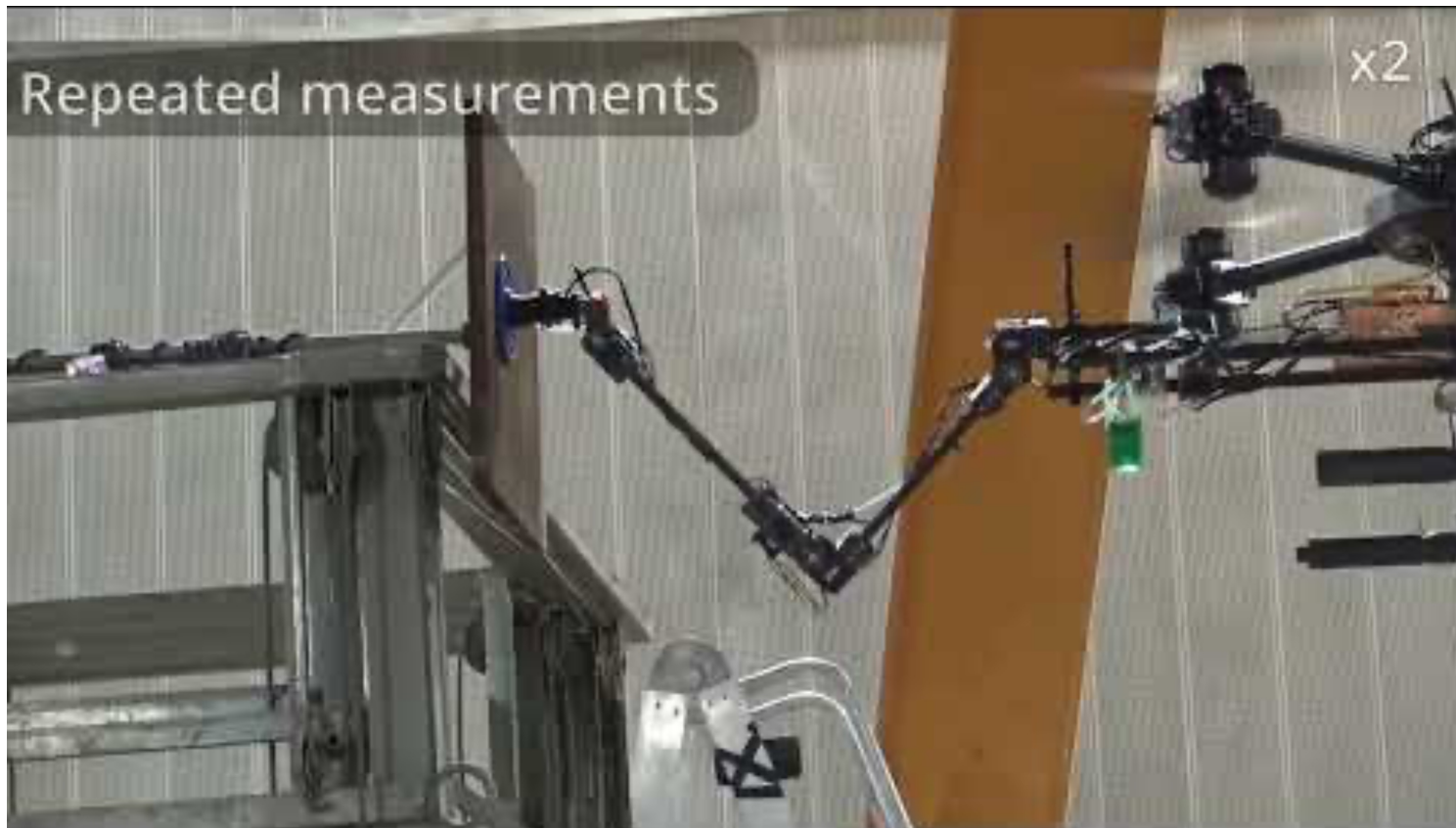
PRISMA Lab

Department of Electrical Engineering and Information Technology

University of Naples Federico II

www.prisma.unina.it

<https://www.youtube.com/watch?v=nffDgxhORdc>



https://www.youtube.com/watch?v=ljDkv3oef_s



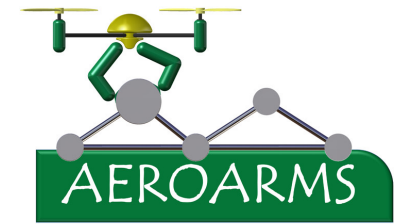
www.aerotrain-etn.eu



www.aerial-core.eu



www.hyfliers.eu



<http://aeroarms-project.eu>



http://cordis.europa.eu/project/rcn/106964_it.html



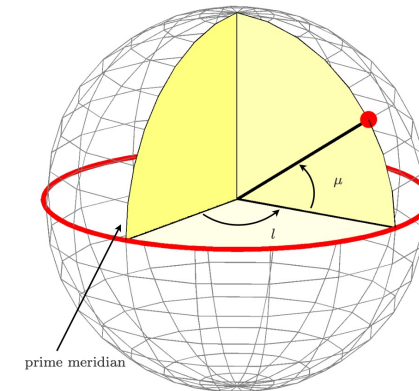
www.arcas-project.eu



<http://airobots.dei.unibo.it>

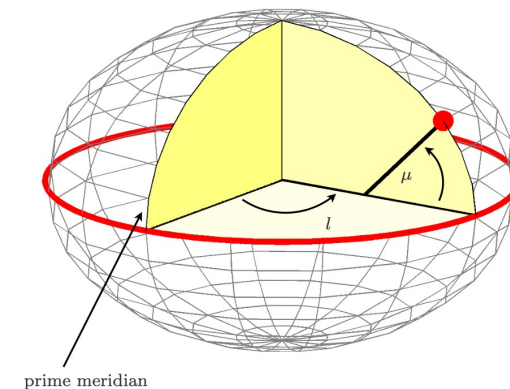
■ Latitude - longitude – altitude

- Latitude: north—south
- Longitude: east – west

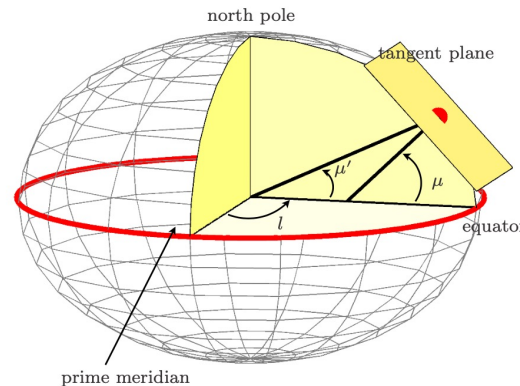


■ Geographic latitude

- Not referred to the earth's centre
 - Except for the poles and points on the prime meridian



■ Geocentric latitude

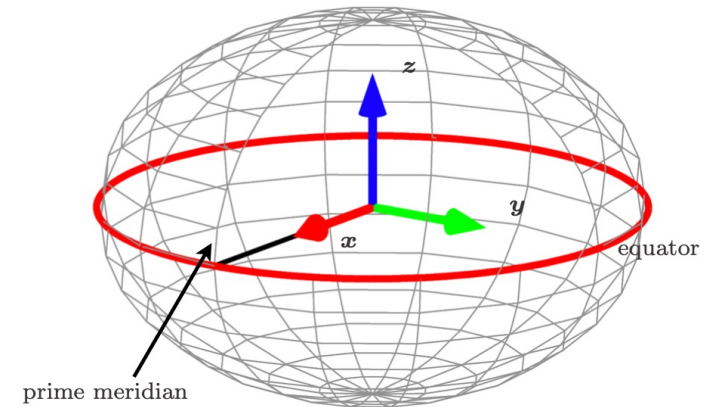


■ Earth-centred inertial frame

- The frame has its origin in the earth's centre
- The orientation is fixed
- The x - y plane lies on the equator plane
- z -axis points to the north pole
- The frame is not subject to any acceleration and it can be considered as inertial

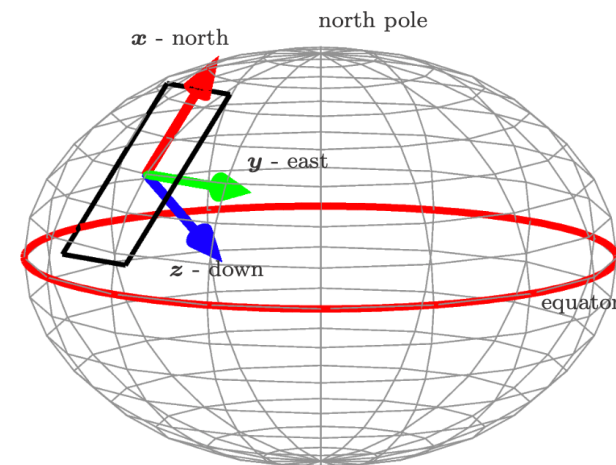


- Earth-centred earth-fixed frame
 - The frame has its origin in the earth's center
 - The orientation follows the earth's rotation
 - The x - y plane lies on the equator plane
 - z -axis points to the north pole
 - The frame is not inertial, but it can be assumed to be so
 - GPS gives positions in this frame



■ North-East-Down (NED) frame

- It is local to the area of interest
- x -axis points to the north
- y -axis points to the east
- z -axis points down
- Opposite to East-North-Up (ENU)



■ Body frame

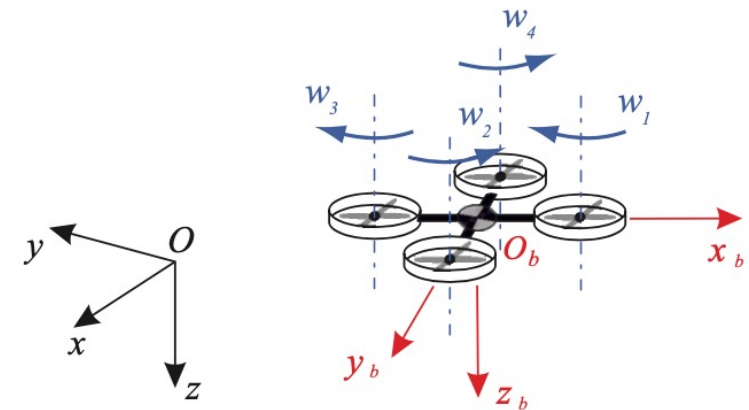
- The origin is at any point of the robot
 - Usually it is located at the center of mass
 - Orientation of the axes is arbitrary
 - It moves together with the robot

■ Frames

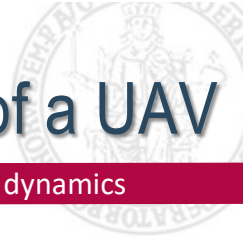
- World frame (North-East-Down configuration)
- Body frame (North-East-Down configuration)

■ Pose

- $p_b = [x \ y \ z]^T \in \mathbb{R}^3$ position
- $R_b = [x_b \ y_b \ z_b] \in SO(3)$ attitude
 - $SO(3)$ is the **special orthogonal group** of dimension three, made by all the three by three rotation matrices
 - The attitude can be also expressed through a minimal representation given by the **ZYX Euler angles (roll-pitch-yaw)**
 - $\eta_b = [\varphi \ \theta \ \psi]^T \in \mathbb{R}^3$



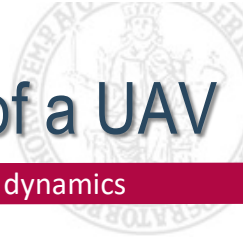
$$R_b(\eta_b) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}$$



- The dynamic model of the UAV can be written in the body frame as

$$\begin{cases} m\ddot{p}_b^b = -mS(\omega_b^b)\dot{p}_b^b + mgR_b^T e_3 + f^b + f_e^b \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b)I_b \omega_b^b + \tau^b + \tau_e^b \end{cases}$$

- $e_3 = [0 \ 0 \ 1]^T$
- $I_b \in \mathbb{R}^{3 \times 3}$ is the diagonal inertia matrix for the angular part and referred to the body frame
- $m > 0$ mass of the UAV
- g gravity acceleration
- $f_e^b, \tau_e^b \in \mathbb{R}^r$ external or unmodelled terms expressed in the body frame
- $f^b = [f_x \ f_y \ f_z]^T \in \mathbb{R}^3$
- $\tau^b = [\tau_x \ \tau_y \ \tau_z]^T \in \mathbb{R}^3$

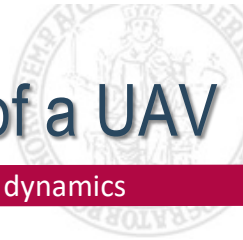


- For many control purposes, the linear part should be expressed in the world frame
- Recall that $\dot{p}_b = R_b \dot{p}_b^b$ and then $\ddot{p}_b = R_b \ddot{p}_b^b + \dot{R}_b \dot{p}_b^b = R_b \ddot{p}_b^b + R_b S(\omega_b^b) \dot{p}_b^b$
 - Multiply both sides of $m \ddot{p}_b^b = -m S(\omega_b^b) \dot{p}_b^b + m g R_b^T e_3 + f^b + f_e^b$ by R_b

$$m R_b \ddot{p}_b^b = -m R_b S(\omega_b^b) \dot{p}_b^b + m g R_b R_b^T e_3 + R_b f^b + R_b f_e^b$$
 - Substituting $R_b \ddot{p}_b^b = \ddot{p}_b - R_b S(\omega_b^b) \dot{p}_b^b$ into the previous one yields

$$m \ddot{p}_b - m R_b S(\omega_b^b) \dot{p}_b^b = -m R_b S(\omega_b^b) \dot{p}_b^b + m g R_b R_b^T e_3 + R_b f^b + R_b f_e^b$$
- Therefore, we obtain the **geometric dynamic model of a UAV**

$$\begin{cases} m \ddot{p}_b = m g e_3 + R_b f^b + f_e \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b) I_b \omega_b^b + \tau^b + \tau_e^b \end{cases}$$



- The dynamic model of the UAV can be written also in terms of the RPY angles as [1]

$$\begin{cases} m\ddot{p}_b = mge_3 + R_b f^b + f_e \\ M(\eta_b)\ddot{\eta}_b = -C(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b + Q^T(\eta_b)\tau_e^b \end{cases}$$

- $M(\eta_b) = Q^T(\eta_b)I_b Q(\eta_b) \in \mathbb{R}^{3 \times 3}$ symmetric and positive definite provided that $\theta \neq \pm \frac{\pi}{2}$
- $C(\eta_b, \dot{\eta}_b) = Q^T(\eta_b)S(Q(\eta_b)\dot{\eta}_b)I_b Q(\eta_b) + Q^T(\eta_b)I_b \dot{Q}(\eta_b) \in \mathbb{R}^{3 \times 3}$ with $S(*) \in \mathbb{R}^{3 \times 3}$ the skew-symmetric operator
- $Q(\eta_b) \in \mathbb{R}^{3 \times 3}$ is the transformation matrix such that $\omega_b^b = Q(\eta_b)\dot{\eta}_b$, with $\omega_b^b \in \mathbb{R}^3$ the angular velocity of the body frame with respect to the world frame expressed in the body frame
- $Q^T(\eta_b)\tau^b$ is an example of kinetostatic duality

$$Q = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & c_\theta s_\phi \\ 0 & -s_\phi & c_\theta c_\phi \end{bmatrix}$$

[1] A. Ollero, B. Siciliano (Eds.), "Aerial robotic manipulation," Springer, 2019.

■ Sketch of the proof

$$\omega_b^b = Q(\eta_b) \dot{\eta}_b \Rightarrow \dot{\eta}_b = Q(\eta_b)^{-1} \omega_b^b, \text{ provided that } \theta \neq \pm \frac{\pi}{2}$$

$$\ddot{\eta}_b = \dot{Q}^{-1} \omega_b^b + Q^{-1} \dot{\omega}_b^b$$

- We replace the coordinate-free quadrotor dynamic model for the angular in $\dot{\omega}_b^b$

$$\ddot{\eta}_b = \dot{Q}^{-1} \omega_b^b + Q^{-1} I_b^{-1} (-S(\omega_b^b) I_b \omega_b^b + \tau^b + \tau_e^b)$$

$$\ddot{\eta}_b = \dot{Q}^{-1} Q \dot{\eta}_b + Q^{-1} I_b^{-1} (-S(Q \dot{\eta}_b) I_b Q \dot{\eta}_b + \tau^b + \tau_e^b)$$

- We multiply both sides of the last expression by $M(\eta_b) = Q^T I_b Q$

$$M(\eta_b) \ddot{\eta}_b = Q^T I_b Q \dot{Q}^{-1} Q \dot{\eta}_b + Q^T I_b Q Q^{-1} I_b^{-1} (-S(Q \dot{\eta}_b) I_b Q \dot{\eta}_b + \tau^b + \tau_e^b)$$

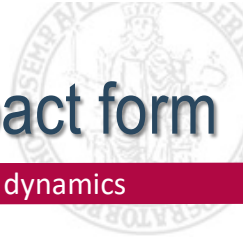
$$M(\eta_b) \ddot{\eta}_b = Q^T I_b Q \dot{Q}^{-1} Q \dot{\eta}_b + Q^T (-S(Q \dot{\eta}_b) I_b Q \dot{\eta}_b + \tau^b + \tau_e^b)$$

- The following expression holds $\dot{Q} = -Q \dot{Q}^{-1} Q \Rightarrow \dot{Q}^{-1} Q = -Q^{-1} \dot{Q}$

- Therefore

$$M(\eta_b) \ddot{\eta}_b = Q^T I_b \dot{Q} \dot{\eta}_b + Q^T (-S(Q \dot{\eta}_b) I_b Q \dot{\eta}_b + \tau^b + \tau_e^b)$$

from which it is possible to have the model seen before



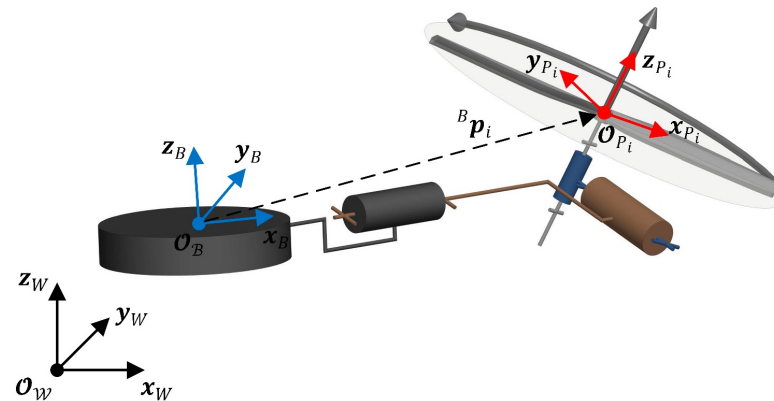
- Stacking $\zeta = [p_b^T \quad \eta_b^T]^T$ from the RPY dynamic model it is possible to obtain the RPY dynamic model of a UAV in a compact form

$$M_\zeta \ddot{\zeta} + C_\zeta \dot{\zeta} + g_\zeta = \Lambda u + \xi_e$$

- $M_\zeta = \begin{bmatrix} mI_3 & O_{3 \times 3} \\ O_{3 \times 3} & M(\eta_b) \end{bmatrix} \in \mathbb{R}^{6 \times 6}$ symmetric and positive definite provided that $\theta \neq \pm \frac{\pi}{2}$
- $C_\zeta = \begin{bmatrix} O_{3 \times 3} & O_{3 \times 3} \\ O_{3 \times 3} & C(\eta_b, \dot{\eta}_b) \end{bmatrix} \in \mathbb{R}^{6 \times 6}$
- $g_\zeta = \begin{bmatrix} mg e_3 \\ 0_3 \end{bmatrix} \in \mathbb{R}^6$
- $\Lambda = \begin{bmatrix} R_b & O_{3 \times 3} \\ O_{3 \times 3} & Q^T(\eta_b) \end{bmatrix} \in \mathbb{R}^{6 \times 6}$ and $u = [f^{b^T} \quad \tau^{b^T}]^T$
 - $\Lambda = \begin{bmatrix} R_b e_3 & O_{3 \times 3} \\ 0_3 & Q^T(\eta_b) \end{bmatrix} \in \mathbb{R}^{6 \times 4}$ and $u = [u_T \quad \tau^{b^T}]^T$ in the quadrotor case
- $\xi_e = \begin{bmatrix} f_e \\ Q^T(\eta_b) \tau_e^b \end{bmatrix} \in \mathbb{R}^6$

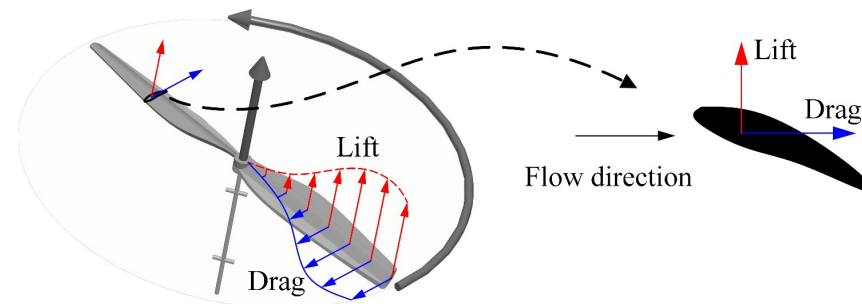
■ Multirotor inputs

- The actuation units are mechatronic components producing thrust
 - They typically consist of a brushless motor with a single propeller attached to the main body either rigidly or through a mechatronic system comprising other joints and servomotors (e.g., tilted/tilting actuation units)



- For each propeller, define a frame rigidly attached to the brushless motor (not spinning with the propeller), whose position w.r.t. the body frame is $p_i^b \in \mathbb{R}^3, i = 1, \dots, n$, and the axis $z_{p_i}^b$ that is parallel to the spinning propeller's axes
 - This frame can change orientation w.r.t. the body frame because of some mechatronic system in between (e.g., tilted/tilting actuation units); in this case the orientation is given by $R_{p_i}^B(\alpha_i, \beta_i) \in SO(3)$, where α_i and β_i are radial and tangential rotation angles

- The propeller rotation produces
 - Thrust force
 - A force that is perpendicular to the air flow direction and to the wing span direction.
 - Drag moment
 - Moment produced by the air resistance, tending to stop the propeller's motion; to keep the propeller to keep the propeller spinning at a certain speed the brushless motor has to apply to the rotor a moment that is opposed to the drag motor. As a byproduct the motor applies on the main body of the multirotor a moment that is equal to the drag moment
- To consider only the thrust force and the drag moment, we neglect
 - The effects due to the propeller inertia (lower inertia of the propeller than the mass and inertia of the motor and the whole multirotor)
 - Flapping effects and other aerodynamic effects



- Model of the thrust force and drag moment

- Far from obstacles and considering a quasi-static analysis, we have

$$f_{P_i} = c_f |\omega_i| \omega_i$$

$$m_{P_i} = k_i c_m |\omega_i| \omega_i$$

- $c_f > 0$, thrust constant; $c_m > 0$, drag factor; $k_i = 1$ descending chord and $k_i = -1$ ascending chord (they are alternated as the rotation direction, producing a thrust always along $z_{P_i}^b$, that is always upward, and to balance the drag moment that otherwise would create a spinning torque around z_b)
- We are considering the magnitude of the force and the moment

- The total thrust force, $f_b \in \mathbb{R}^3$, and drag moment, $\tau_b \in \mathbb{R}^3$, produced by the propellers on the multirotor w.r.t. the body frame are the net force and moment produced by each propeller as

$$f^b = \sum_{i=1}^n c_f |\omega_i| \omega_i z_{P_i}^b$$

$$\tau^b = \sum_{i=1}^n |\omega_i| \omega_i (-k_i c_m z_{P_i}^b + c_f S(p_{P_i}^b) z_{P_i}^b)$$

- Besides, assuming that the brushless motor are velocity controlled, it is possible to define $u_{\omega_i} = |\omega_i| \omega_i \in \mathbb{R}$ as one of the controllable input for the i -th propeller

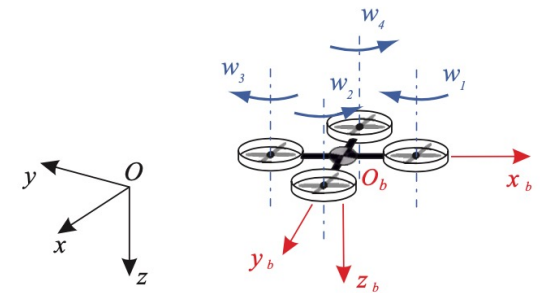
- Therefore, it is possible to write

$$\begin{bmatrix} f^b \\ \tau^b \end{bmatrix} = f_u(\alpha_i, \beta_i, u_{\omega_i})$$

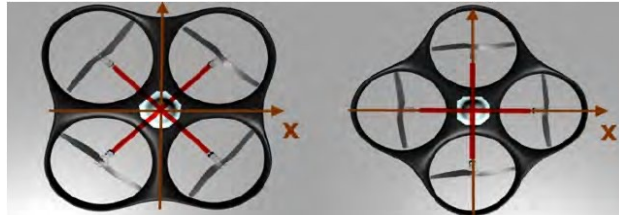
- For a quadrotor, one has $\alpha_i = 0, \beta_i = 0, z_{P_i}^b = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$, therefore

$$\begin{bmatrix} f_x \\ f_y \\ f_z \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -c_f & -c_f & -c_f & -c_f \\ 0 & -lc_f & 0 & lc_f \\ lc_f & 0 & -lc_f & 0 \\ -c_m & c_m & -c_m & c_m \end{bmatrix}}_{\text{Allocation matrix}} \begin{bmatrix} u_{\omega_1} \\ u_{\omega_2} \\ u_{\omega_3} \\ u_{\omega_4} \end{bmatrix} = G_q u_{\omega}$$

- With $f_z = u_T = \sum_{i=1}^4 f_{P_i} > 0$



- The allocation matrix plays the role of the input matrix into the dynamic model
- It is clear that the quadrotor is underactuated due to the presence of two zero rows
- The horizontal movements are controlled by directing the total thrust vector $u_T z_b$ in the appropriate direction (**thrust vectoring control**), thereby resulting in longitudinal and lateral force components
- The allocation matrix for a UAV with the propellers in a flat configuration is $G_q \in \mathbb{R}^{6 \times n}$, with n the number of propellers
 - Notice that, if $n > 4$, but the propellers are co-planar, you have a redundancy in how to split the velocities among all the propellers, but the system is still underactuated since the n co-planar propellers do not give you the possibility to have more real inputs to the system than 4
 - To get $G_q \in \mathbb{R}^{6 \times n}$ and avoid the underactuation the UAV must have either non co-planar propellers (**tilted configurations**) or motorised propellers (**actively tilting configurations**)
- The allocation matrix is invertible for quadrotors but it should be checked for different configurations



■ Dynamic models for the particular case of the quadrotor

$$\begin{cases} m\ddot{p}_b^b = -mS(\omega_b^b)\dot{p}_b^b + mgR_b^T e_3 - u_T e_3 + f_e^b \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b)I_b \omega_b^b + \tau^b + \tau_e^b \end{cases}$$

$$\begin{cases} m\ddot{p}_b^b = mge_3 - u_T R_b e_3 + f_e \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b)I_b \omega_b^b + \tau^b + \tau_e^b \end{cases}$$

$$\begin{cases} m\ddot{p}_b = mge_3 - u_T R_b e_3 + f_e \\ M(\eta_b)\ddot{\eta}_b = -C(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b + Q^T(\eta_b)\tau_e^b \end{cases}$$



- The quadrotor is a differentially flat system
 - (x, y, z, ψ) are the flat outputs
- Therefore, it is reasonable to plan the UAV path/time law on the basis of the flat outputs
- Notice that the full attitude of the UAV cannot be specified, but it depends on the position and the yaw and their time derivatives
 - This is due to the system underactuation
 - To obtain the desired free attitude, other UAVs as the tilting propellers drones must be employed



- For the quadrotor described by the RPY dynamic model, the state is $(\zeta, \dot{\zeta})$
- The flat outputs are

$$\sigma = \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \\ \psi \end{bmatrix}$$

- The relation between the state and the flat outputs is

$$\begin{aligned} x &= \sigma_1 \\ y &= \sigma_2 \\ z &= \sigma_3 \\ \phi &= \sin^{-1} \left(\frac{\ddot{\sigma}_2 \cos \sigma_4 - \ddot{\sigma}_1 \sin \sigma_4}{\sqrt{\ddot{\sigma}_1^2 + \ddot{\sigma}_2^2 + (\ddot{\sigma}_3 - g)^2}} \right) \\ \theta &= \tan^{-1} \left(\frac{\ddot{\sigma}_1 \cos \sigma_4 + \ddot{\sigma}_2 \sin \sigma_4}{\ddot{\sigma}_3 - g} \right) \\ \psi &= \sigma_4 \end{aligned}$$



- Then, for the velocity

$$\dot{x} = \dot{\sigma}_1$$

$$\dot{y} = \dot{\sigma}_2$$

$$\dot{z} = \dot{\sigma}_3$$

$$\dot{\psi} = \dot{\sigma}_4$$

- For the roll and pitch velocity, recall $\dot{\eta}_b = Q^{-1}\omega_b^b$ and $\dot{R}_b = R_b S(\omega_b^b) \Rightarrow S(\omega_b^b) = R_b^T \dot{R}_b$

- The inverse of the skew-symmetric operator is the **vee operator**

$$\omega_b^b = S(\omega_b^b)^V$$

- Example

$$\omega_b^b = \begin{bmatrix} \omega_{b,x}^b \\ \omega_{b,y}^b \\ \omega_{b,z}^b \end{bmatrix} \text{ from which } S(\omega_b^b) = \begin{bmatrix} 0 & -\omega_{b,z}^b & \omega_{b,y}^b \\ \omega_{b,z}^b & 0 & -\omega_{b,x}^b \\ -\omega_{b,y}^b & \omega_{b,x}^b & 0 \end{bmatrix} \text{ therefore if } S(\omega_b^b) = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & 4 \\ 2 & -4 & 0 \end{bmatrix} \text{ then } \omega_b^b = \begin{bmatrix} -4 \\ -2 \\ 1 \end{bmatrix}$$

- Then, $\dot{\eta}_b = Q^{-1}(R_b^T \dot{R}_b)^V$ by taking the RPY angles from the above expressions



- Finally, for the inputs

$$u_T = m \sqrt{\ddot{\sigma}_1^2 + \ddot{\sigma}_2^2 + (\ddot{\sigma}_3 - g)^2}$$

$$\tau_b^b = I_b (\dot{R}_b^T + R_b^T \ddot{R}_b)^V + R_b^T \dot{R}_b I_b (R_b^T \dot{R}_b)^V$$

- Therefore, it is reasonable to plan a UAV path/time law on the basis of its flat outputs
- Notice that the dynamics is included within the inputs
 - Therefore, these are valid control inputs as long as we are in ideal conditions and only feed-forward can be sufficient
- It is not possible to plan in advance the quadrotor's attitude, that is instead related to the position
 - This is due because of the underactuation of the quadrotor
 - To obtain the desired free attitude, other UAVs must be considered (tilted or tilting propellers drone)



- We start from the RPY quadcopter dynamic model, neglecting external disturbances

$$\begin{cases} m\ddot{p}_b = mge_3 - u_T R_b e_3 & \text{(1)} \\ M(\eta_b)\ddot{\eta}_b = -C(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b \end{cases}$$

- The controller's objective is to obtain two linear subsystems (one for the linear and the other for the angular part) coupled by a non-linear interconnection term
- To this purpose, notice that the angular part is feedback linearizable

$$\tau^b = I_b Q(\eta_b)\tilde{\tau} + Q(\eta_b)^{-T} C(\eta_b, \dot{\eta}_b)\dot{\eta}_b$$

- $\tilde{\tau} \in \mathbb{R}^3$ is a virtual control input
 - Singularity when $\theta = \pm \frac{\pi}{2}$
- Substituting this control-law within the angular part of the RPY quadcopter dynamic model yields

$$\ddot{\eta}_b = \tilde{\tau}$$

- $M(\eta_b) = Q^T(\eta_b)I_b Q(\eta_b)$ has been exploited



- Consider the tracking error for the attitude

$$\begin{aligned} e_\eta &= \begin{bmatrix} e_\phi \\ e_\theta \\ e_\psi \end{bmatrix} = \eta_b - \eta_{b,d} = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix} - \begin{bmatrix} \phi_d \\ \theta_d \\ \psi_d \end{bmatrix} \\ \dot{e}_\eta &= \begin{bmatrix} \dot{e}_\phi \\ \dot{e}_\theta \\ \dot{e}_\psi \end{bmatrix} = \dot{\eta}_b - \dot{\eta}_{b,d} = \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} - \begin{bmatrix} \dot{\phi}_d \\ \dot{\theta}_d \\ \dot{\psi}_d \end{bmatrix} \end{aligned} \quad (2)$$

- $\eta_{b,d}, \dot{\eta}_{b,d}, \ddot{\eta}_{b,d} \in \mathbb{R}^3$ the desired attitude, its desired velocity and acceleration, respectively
 - How to get the desired orientation if the planner gives us only the desired yaw? Wait some slides, now suppose to have it (!)
- Consider the following definition, that is is like the virtual desired acceleration oriented as the desired attitude, regulated by the total thrust, plus gravity

$$\mu_d = [\mu_x \quad \mu_y \quad \mu_z]^T = -\frac{1}{m} u_T R_b(\eta_{b,d}) e_3 + g e_3 \in \mathbb{R}^3$$

- From (2), we have

$$\eta_b = e_\eta + \eta_{b,d}$$

- That is $\phi = e_\phi + \phi_d$, $\theta = e_\theta + \theta_d$, and $\psi = e_\psi + \psi_d$
- We can substitute this expression into $R_b(\eta_b)$ of (1) in the previous slide

$$(3) \begin{cases} \ddot{p}_b = \mu_d + \frac{1}{m} u_T \delta(\eta_{b,d}, e_\eta) \\ \ddot{\eta}_b = \tilde{\tau} \end{cases}$$

- $\delta(\eta_{b,d}, e_\eta)$ is the interconnection term between the linear and the angular part

$$\delta(\eta_{b,d}, e_\eta) = \begin{bmatrix} s_{\phi_d} s_{\psi_d} - s_{\phi} s_{\psi} - c_{\phi} c_{\psi} s_{\theta} + c_{\phi_d} c_{\psi_d} s_{\theta_d} \\ c_{\psi} s_{\theta} - s_{\phi_d} c_{\psi_d} + c_{\phi_d} s_{\psi_d} s_{\theta_d} - c_{\phi} c_{\psi} s_{\theta} \\ c_{\phi_d} c_{\theta_d} - c_{\phi} c_{\theta} \end{bmatrix}$$

- The above expressions have been obtained exploiting this property

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

- Now, forgetting for one moment the underactuation, we consider μ_d and $\tilde{\tau}$ our inputs in (3)



- Consider the following definitions

$$e_p = p_b - p_{b,d}$$

$$\dot{e}_p = \dot{p}_b - \dot{p}_{b,d}$$

- $p_{b,d}, \dot{p}_{b,d}, \ddot{p}_{b,d} \in \mathbb{R}^3$ the desired position, its desired velocity and acceleration, respectively

- It is thus possible to rewrite (3) as follows

$$(4) \quad \begin{cases} \begin{bmatrix} \dot{e}_p \\ \ddot{e}_p \end{bmatrix} = A \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} + B(\mu_d - \ddot{p}_{b,d}) + \frac{1}{m}u_T\Delta \\ \begin{bmatrix} \dot{e}_\eta \\ \ddot{e}_\eta \end{bmatrix} = A \begin{bmatrix} e_\eta \\ \dot{e}_\eta \end{bmatrix} + B(\tilde{\tau} - \ddot{\eta}_{b,d}) \end{cases}$$

- with

$$A = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{6 \times 6}, B = \begin{bmatrix} 0 \\ I \end{bmatrix} \in \mathbb{R}^{6 \times 3}, \Delta = \begin{bmatrix} 0 \\ \delta(\eta_{b,d}, e_\eta) \end{bmatrix} \in \mathbb{R}^6$$

- As said, the inputs for this system are μ_d and $\tilde{\tau}$ that must bring to zero the error for the linear and angular parts



- We can design the following controller

$$(5) \quad \begin{cases} \mu_d = -K_p \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} + \ddot{p}_{b,d} \\ \tilde{\tau} = -K_e \begin{bmatrix} e_\eta \\ \dot{e}_\eta \end{bmatrix} + \ddot{\eta}_{b,d} \end{cases}$$

- $K_p \in \mathbb{R}^{3 \times 6}$ chosen such that $A_p = A - BK_p$ is Hurwitz
 - $K_e \in \mathbb{R}^{3 \times 6}$ chosen such that $A_e = A - BK_e$ is Hurwitz
- Substituting (5) into (4) yields the following loop equations

$$\begin{cases} \begin{bmatrix} \dot{e}_p \\ \ddot{e}_p \end{bmatrix} = A_p \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} + \frac{1}{m} u_T \Delta \\ \begin{bmatrix} \dot{e}_\eta \\ \ddot{e}_\eta \end{bmatrix} = A_e \begin{bmatrix} e_\eta \\ \dot{e}_\eta \end{bmatrix} \end{cases}$$

- To show the asymptotic stability of the errors equal to zero, the interconnection term requires the use of the perturbation theory



- Now, we have to solve the pending questions
 - How is it possible to compute the full desired attitude and its time derivatives if the planner give us only the desired position and the desired yaw?
 - How is it possible to compute the desired thrust from μ_d ?
- The solution comes from the definition of the flat outputs

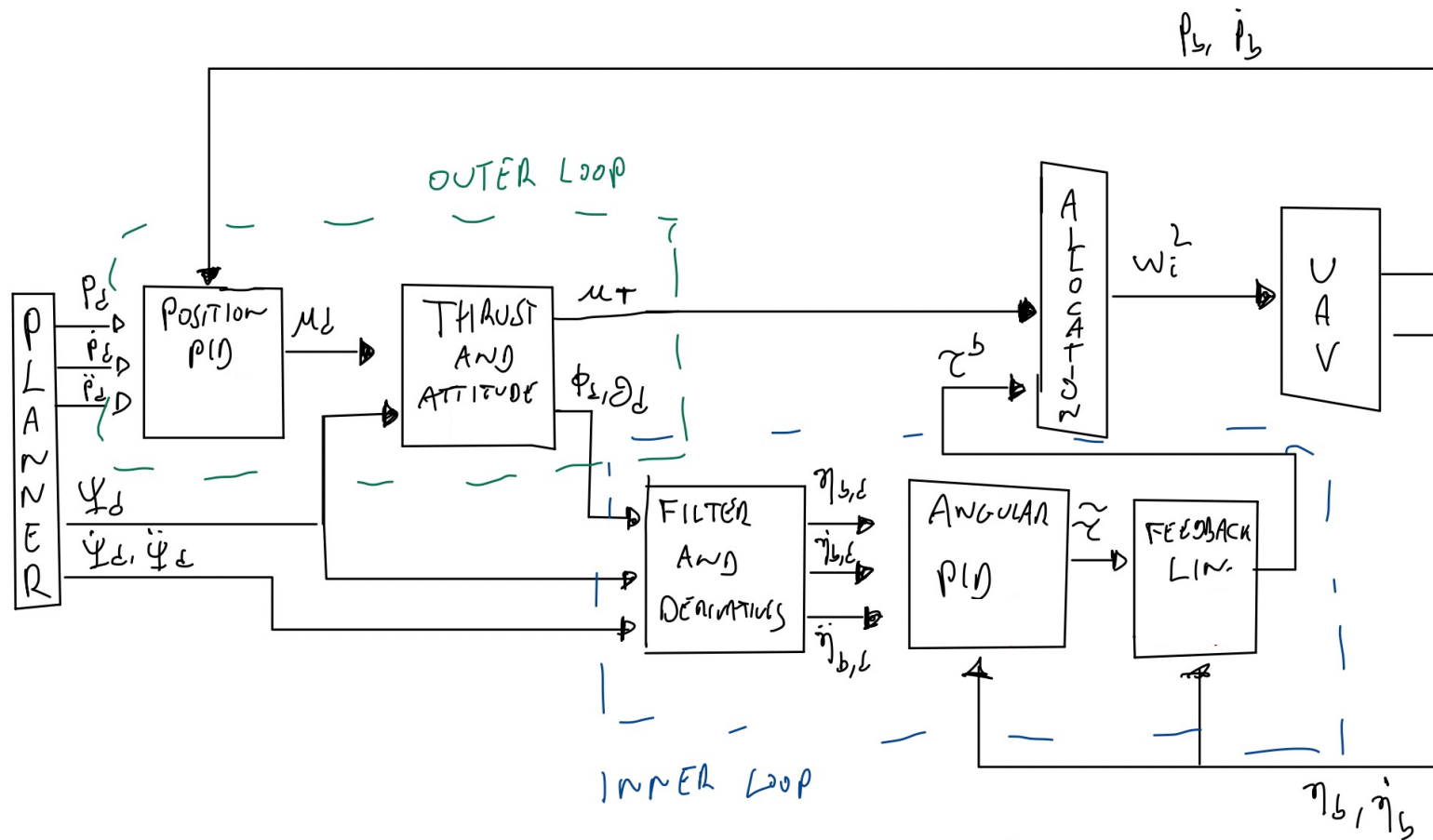
$$\begin{aligned}
 u_T &= m \sqrt{\mu_x^2 + \mu_y^2 + (\mu_z - g)^2} \\
 \varphi_d &= \sin^{-1} \left(\frac{m}{u_T} (\mu_y \cos \psi_d - \mu_x \sin \psi_d) \right) \\
 \theta_d &= \tan^{-1} \left(\frac{\mu_x \cos \psi_d + \mu_y \sin \psi_d}{\mu_z - g} \right)
 \end{aligned}
 \tag{6}$$

- Notice that there is a singularity when $\mu_z = g$, therefore the planner should be designed such that $|\mu_z| < g$



- Outer loop
 - Given the desired linear position, velocity, and acceleration from the planner, it is possible to compute $\mu_d = -K_p \begin{bmatrix} e_p \\ \dot{e}_p \end{bmatrix} + \ddot{p}_{b,d}$
 - This control law can be modified by adding an integral action of the position error without destroying the asymptotic stability property
 - This is needed to add robustness and increase tracking accuracy
 - Then, we find the total thrust and the reference for the roll and the pitch angles
- Inner loop
 - Notice that (6) gives only the roll and the pitch angles
 - In order to retrieve the desired angular velocity and acceleration, we must numerically derivate these values
 - Therefore, a second-order low-pass digital filter should be employed to both reduce noise and compute the time derivatives of such angles
 - After $\tilde{\tau}$, it is possible to compute τ^b from the initial feedback linearization
 - Having the total thrust and τ^b , it is then possible to compute the propellers' velocities
- Considerations
 - Such a two-loop architecture is said to be **hierarchical** [2]
 - It is also justified because the angular dynamics are faster than linear ones (**time-scale separation**)
 - The controller is not suitable for acrobatic flights since $\theta \neq \pm \frac{\pi}{2}$ to avoid singularities

[2] K. Nonami, F. Kendoul, S. Suzuki, W. Wang, D. Nakazawa, "Autonomous flying robots. Unmanned aerial vehicles and micro aerial vehicles," Springer, 2010





- We start from the coordinate-free quadcopter dynamic model, neglecting external disturbances [3]

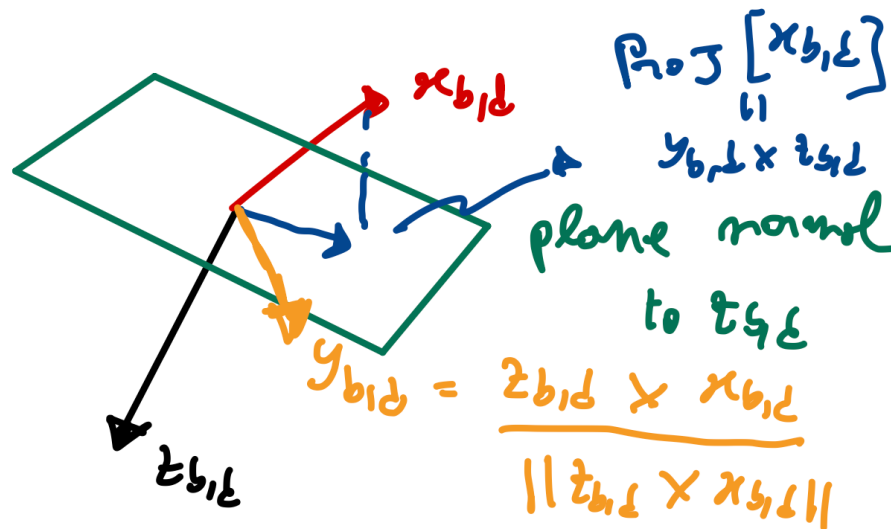
$$\begin{cases} m\ddot{p}_b = mge_3 - u_T R_b e_3 \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b) I_b \omega_b^b + \tau^b \end{cases}$$

- As in the hierarchical controller, the translational motion is only dependent on the term $u_T R_b e_3$
 - The magnitude of the total thrust, u_T , is directly controlled, but the direction of the total thrust $R_b e_3$ is determined by the third body-fixed axis
 - To change the direction of the total thrust, the attitude should be changed accordingly
- We choose the total thrust, u_T , and the desired reduced attitude, $R_{b,d} e_3$, for the third body-fixed axis such that they stabilize the zero equilibrium of the tracking error for the translational dynamics
- The remaining first two columns of $R_{b,d}$, representing the direction of the body-fixed axes, are chosen to follow the desired direction
 - The control torque, τ^b , is designed to follow the resulting desired attitude $R_{b,d}$ obtained by $x_{b,d}$ and $z_{b,d} = R_{b,d} e_3$

[3] T. Lee, M. Leok, N.H. McClamroch, "Geometric tracking control of quadrotor UAV on SE(3)," 49th IEEE Conference on Decision and Control, 2010



- Once $z_{b,d} = R_{b,d}e_3$ is defined, the desired $x_{b,d}$ might not be orthogonal to it
 - $x_{b,d}$ must be project into the plane orthogonal to $z_{b,d}$



$$R_{b,d} = \begin{bmatrix} S(y_{b,d})z_{b,d} & \underbrace{\frac{S(z_{b,d})x_{b,d}}{\|S(z_{b,d})x_{b,d}\|}}_{y_{b,d}} & z_{b,d} \end{bmatrix}$$



- Controller's job

- $p_b \rightarrow p_{b,d}$
- $\dot{p}_b \rightarrow \dot{p}_{b,d}$
- $Proj[x_b] \rightarrow Proj[x_{b,d}]$

- Tracking errors for the translational motion

$$e_p = p_b - p_{b,d}, \dot{e}_p = \dot{p}_b - \dot{p}_{b,d}$$

- Tracking errors for the angular motion

$$e_R = \frac{1}{2} (R_{b,d}^T R_b - R_b^T R_{b,d})^\vee$$

$$e_\omega = \omega_b^b - R_b^T R_{b,d} \omega_{b,d}^{b,d}$$

- The angular error is defined in $SO(3)$
- The angular velocity error can be seen as the velocity of $R_{b,d}^T R_b$ in the body frame (see [3])



- Assumption

$$\| -mge_3 + m\ddot{p}_{b,d} \| < \text{positive constant}$$

- This ensures that the planned desired linear acceleration does not exceed the gravity acceleration

- Outer loop control

$$u_T = -(-K_p e_p - K_v \dot{e}_p - mge_3 + m\ddot{p}_{b,d})^T R_b e_3$$

$$z_{b,d} = - \frac{-K_p e_p - K_v \dot{e}_p - mge_3 + m\ddot{p}_{b,d}}{\| -K_p e_p - K_v \dot{e}_p - mge_3 + m\ddot{p}_{b,d} \|}$$

- Inner loop control (after retrieved $R_{b,d}$)

$$\tau^b = -K_R e_R - K_\omega e_\omega + S(\omega_b^b) I_b \omega_b^b - I_b (S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d} - R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d})$$



- Closed-loop system

$$m\ddot{e}_p + K_v\dot{e}_p + K_p e_p = - \underbrace{\frac{u_T}{e_3^T R_{b,d}^T R_b e_3} [(e_3^T R_{b,d}^T R_b e_3) R_b e_3 - R_{b,d} e_3]}_X$$

Plays the role of the interconnection term as seen in the hierarchical control, and it is zero when the orientation error is zero

$$I_b \dot{e}_\omega + K_\omega e_\omega + K_R e_R = 0$$

- Notice that only Lyapunov theory can give us the exponential convergence of the error to zero
 - This is true if the initial attitude error is less than 90°



- Sketch of how to obtain the closed-loop system (angular part)

- Recall the definition of the angular velocity error

$$e_\omega = \omega_b^b - R_b^T R_{b,d} \omega_{b,d}^{b,d}$$

- Taking its time derivative yields

$$\dot{e}_\omega = \dot{\omega}_b^b - \frac{d}{dt}(R_b^T R_{b,d} \omega_{b,d}^{b,d})$$

- Multiply both sides by I_b , we have $I_b \dot{e}_\omega = I_b \dot{\omega}_b^b - I_b \frac{d}{dt}(R_b^T R_{b,d} \omega_{b,d}^{b,d})$
- Substituting the system angular dynamics in the purple term above yields

$$I_b \dot{e}_\omega = -S(\omega_b^b) I_b \omega_b^b + \tau^b - I_b \frac{d}{dt}(R_b^T R_{b,d} \omega_{b,d}^{b,d})$$

- Notice that

$$\frac{d}{dt}(R_b^T R_{b,d} \omega_{b,d}^{b,d}) = R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d} + \dot{R}_b^T R_{b,d} \omega_{b,d}^{b,d} + R_b^T \dot{R}_{b,d} \omega_{b,d}^{b,d}$$

- Recalling that $\dot{R}_{b,d} = R_{b,d} S(\omega_{b,d}^{b,d})$, then $\dot{R}_b^T = -S(\omega_b^b) R_b^T$, then

$$\frac{d}{dt}(R_b^T R_{b,d} \omega_{b,d}^{b,d}) = R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d} - \underbrace{S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d}}_0 + R_b^T R_{b,d} S(\omega_{b,d}^{b,d}) \omega_{b,d}^{b,d}$$

- Finally

$$I_b \dot{e}_\omega = -S(\omega_b^b) I_b \omega_b^b + \tau^b - I_b (R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d} - S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d})$$

- Substituting $\tau^b = -K_R e_R - K_\omega e_\omega + S(\omega_b^b) I_b \dot{\omega}_b^b - I_b (S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d} - R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d})$ into the previous expression yields

$$\begin{aligned}
 I_b \dot{e}_\omega &= -S(\omega_b^b) I_b \dot{\omega}_b^b + -K_R e_R - K_\omega e_\omega + S(\omega_b^b) I_b \dot{\omega}_b^b - I_b (S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d} - R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d}) \\
 &\quad - I_b (R_b^T R_{b,d} \dot{\omega}_{b,d}^{b,d} - S(\omega_b^b) R_b^T R_{b,d} \omega_{b,d}^{b,d})
 \end{aligned}$$

- We then obtain $I_b \dot{e}_\omega + K_\omega e_\omega + K_R e_R = 0$
- Since e_ω is **not** the time derivative of e_R , the proof of exponential stability for the above closed-loop system is obtained from Lyapunov theory

- Sketch of how to obtain the closed-loop system (linear part)
 - Recall the definition of the position velocity error $\dot{e}_p = \dot{p}_b - \dot{p}_{b,d}$, it is possible to write $\ddot{e}_p = \ddot{p}_b - \ddot{p}_{b,d}$
 - Multiply both sides by the mass of the quadcopter, $m\ddot{e}_p = m\ddot{p}_b - m\ddot{p}_{b,d}$
 - Substituting the system linear dynamics in the purple term above yields

$$m\ddot{e}_p = mge_3 - u_T R_b e_3 - m\ddot{p}_{b,d}$$

- Add and subtract the term $\frac{u_T}{e_3^T R_{b,d}^T R_b e_3} R_{b,d} e_3$

$$m\ddot{e}_p = mge_3 - u_T R_b e_3 - m\ddot{p}_{b,d} + \frac{u_T}{e_3^T R_{b,d}^T R_b e_3} R_{b,d} e_3 - \frac{u_T}{e_3^T R_{b,d}^T R_b e_3} R_{b,d} e_3$$

- $R_b e_3 = z_b$
- $e_3^T R_{b,d}^T = z_{b,d}^T$
- $e_3^T R_{b,d}^T R_b e_3$ is the cosine angle between z_b and $z_{b,d}$

- Recall the term $X = \frac{u_T}{e_3^T R_{b,d}^T R_b e_3} [(e_3^T R_{b,d}^T R_b e_3) R_b e_3 - R_{b,d} e_3]$, the previous expression can be written as

$$m\ddot{e}_p = mge_3 - m\ddot{p}_{b,d} - \frac{u_T}{e_3^T R_{b,d}^T R_b e_3} R_{b,d} e_3 - X$$

- Let $A = -K_p e_p - K_v \dot{e}_p - mge_3 + m\ddot{p}_{b,d}$ be the desired control force for the translational dynamics, then the total thrust should be taken as $u_T = -A^T R_b e_3$
- However, $A = \|A\| R_{b,d} e_3$, since it is directed along the z -axis of the desired body frame, and therefore $u_T = -\|A\| (R_{b,d} e_3)^T R_b e_3$
- Substituting the total thrust into the above expression yields

$$m\ddot{e}_p = mge_3 - m\ddot{p}_{b,d} + \frac{\|A\| (R_{b,d} e_3)^T R_b e_3}{e_3^T R_{b,d}^T R_b e_3} \frac{A}{\|A\|} - X$$

- We get

$$m\ddot{e}_p = mge_3 - m\ddot{p}_{b,d} + A - X$$

- That is

$$m\ddot{e}_p = \cancel{mge_3 - m\ddot{p}_{b,d}} - K_p e_p - K_v \dot{e}_p = \cancel{mge_3 - m\ddot{p}_{b,d}} - X$$

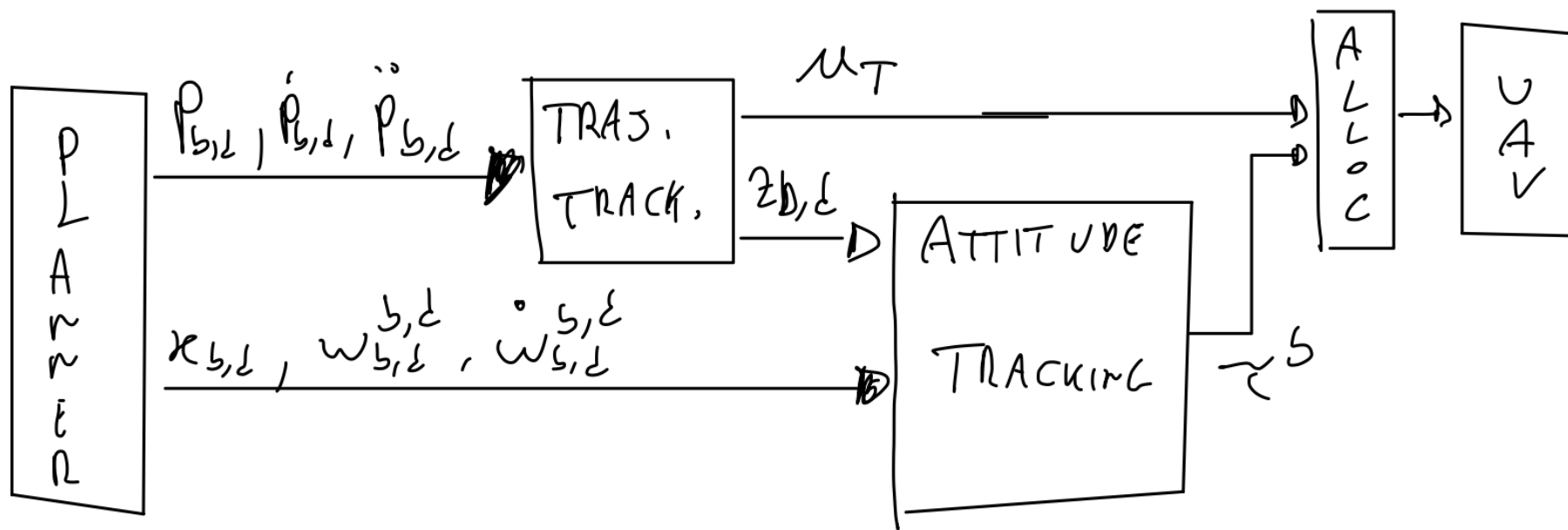
- We then obtain $m\ddot{e}_p + K_v \dot{e}_p + K_p e_p = -X$
- The interconnection term X plays the role of the interconnection term already seen within the hierarchical controller
- Notice that if $R_{b,d} = R_b$, then $X = 0$
- Lyapunov and perturbation theory give us conditions on the boundness of $\|e_p\|$ and $\|\dot{e}_p\|$

■ Properties

- One of the unique properties of the presented geometric controller is that it is entirely designed in the model-configuration space $\mathbb{R}^3 \times SO(3)$ using rotation matrices, avoiding any representation singularities
- Exponential stability of the closed-loop system can be proven if the initial attitude error is less than 90°
- The total thrust can be written as

$$u_T = -\|A\| (R_{b,d} e_3)^T R_b e_3 = -\|A\| z_{b,d}^T z_b$$

- It contains the scalar product between $z_{b,d}$ and z_b
 - If we have a big error in the attitude, this results in a big angle between these two axes; however, in this case, the total thrust decreases in magnitude since the dot product decreases at big angles
- Therefore, this controller is designed such that the position tracking error converges to zero when there is no attitude error, and it is limited for non-zero attitude tracking errors to achieve asymptotic stability of the complete closed-loop dynamics





- Consider the RPY quadrotor dynamic model and, now, external wrench disturbance

$$\begin{cases} m\ddot{p}_b = mge_3 - u_TR_be_3 + f_e \\ M(\eta_b)\ddot{\eta}_b = -C(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b + \tau_e \end{cases}$$

- Consider the generalised momentum vector $q \in \mathbb{R}^6$

$$q = \begin{bmatrix} mI_3 & O_3 \\ O_3 & M(\eta_b) \end{bmatrix} \begin{bmatrix} \dot{p}_b \\ \dot{\eta}_b \end{bmatrix}$$

- I_X and O_X are identity and zero matrices of proper dimensions

- The time derivative of the momentum is

$$\dot{q} = \begin{bmatrix} mge_3 - u_TR_be_3 + f_e \\ C^T(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b + \tau_e \end{bmatrix}$$

- The notable property $\dot{M}(\eta_b) = C^T(\eta_b, \dot{\eta}_b) + C(\eta_b, \dot{\eta}_b)$ has been exploited



- Estimator's goal
 - Estimate the wrench $\begin{bmatrix} f_e \\ \tau_e \end{bmatrix}$ through $\begin{bmatrix} \hat{f}_e \\ \hat{\tau}_e \end{bmatrix}$
- To simplify things, we want a linear relationship between the external wrench and its estimation in the Laplace domain

$$\mathcal{L} \left[\begin{bmatrix} \hat{f}_e \\ \hat{\tau}_e \end{bmatrix} \right] = G(s) \mathcal{L} \left[\begin{bmatrix} f_e \\ \tau_e \end{bmatrix} \right]$$

- $G(s) \in \mathbb{C}^{6 \times 6}$ is a diagonal matrix of transfer functions
- To further simplify, let's choose first-order transfer functions

$$G_i(s) = \frac{k_o}{s + c_o}, i = 1, \dots, 6$$

- $k_o, c_o > 0$
 - What is the best choice for the gains?
 - $c_o \rightarrow +\infty$... it is impossible: a compromise must be found
 - $k_o = c_o$ not to have any gain in the estimation



- Apply the inverse of the Laplace operator to $\mathcal{L} \begin{bmatrix} \hat{f}_e \\ \hat{\tau}_e \end{bmatrix} = G(s) \mathcal{L} \begin{bmatrix} f_e \\ \tau_e \end{bmatrix}$, with the hypotheses that

$$\begin{bmatrix} \hat{f}_e(0) \\ \hat{\tau}_e(0) \end{bmatrix} = 0 \text{ and } q(0) = 0$$

- This means that the estimation should start before the UAV's take-off

$$\frac{d}{dt} \begin{bmatrix} \hat{f}_e(t) \\ \hat{\tau}_e(t) \end{bmatrix} = k_o \begin{bmatrix} f_e(t) \\ \tau_e(t) \end{bmatrix} - k_o \begin{bmatrix} \hat{f}_e(t) \\ \hat{\tau}_e(t) \end{bmatrix}$$

- Recall

$$\dot{q} = \begin{bmatrix} mge_3 - u_T R_b e_3 + f_e \\ C^T(\eta_b, \dot{\eta}_b) \dot{\eta}_b + Q^T(\eta_b) \tau^b + \tau_e \end{bmatrix} \Rightarrow \begin{bmatrix} f_e \\ \tau_e \end{bmatrix} = \dot{q} - \begin{bmatrix} mge_3 - u_T R_b e_3 \\ C^T(\eta_b, \dot{\eta}_b) \dot{\eta}_b + Q^T(\eta_b) \tau^b \end{bmatrix}$$



- Therefore, folding the last expression into the purple equation before and integrating yield

$$\begin{bmatrix} \hat{f}_e \\ \hat{t}_e \end{bmatrix} = k_o \left(q - \int_0^t \begin{bmatrix} \hat{f}_e \\ \hat{t}_e \end{bmatrix} + \begin{bmatrix} mge_3 - u_T R_b e_3 \\ C^T(\eta_b, \dot{\eta}_b) \dot{\eta}_b + Q^T(\eta_b) \tau^b \end{bmatrix} dt \right)$$

- The necessary quantities to estimate the external wrench are
 - Attitude → IMU
 - Angular velocity → IMU
 - Linear velocity → GPS/on-board devices
 - Inputs → Employed controller
 - Model parameters → System modelling
- The integral of the estimator is approximated numerically
- The estimation can be further filtered (e.g., Butterworth filters)
 - Higher-order transfer functions can be used
- The same estimator can be obtained from the coordinate-free dynamic model



- How can we use the estimation inside the hierarchical controller?
- We can do the following modifications to the equations above
 - Partial feedback linearization

$$\tau^b = I_b Q(\eta_b) \tilde{\tau} + Q^{-T}(\eta_b) C(\eta_b, \dot{\eta}_b) \dot{\eta}_b - Q^{-T}(\eta_b) \hat{\tau}_e$$

- Definition

$$\mu_d = [\mu_x \quad \mu_y \quad \mu_z]^T = -\frac{1}{m} u_T R_b(\eta_b) e_3 + g e_3 + \frac{1}{m} \hat{f}_e$$

- Thrust and desired roll-pitch computation

$$\bar{\mu} = [\bar{\mu}_x \quad \bar{\mu}_y \quad \bar{\mu}_z]^T = \mu - \frac{1}{m} \hat{f}_e$$

$$u_T = m \sqrt{\bar{\mu}_x^2 + \bar{\mu}_y^2 + (\bar{\mu}_z - g)^2}$$

$$\varphi_d = \sin^{-1} \left(\frac{m}{u_T} (\bar{\mu}_y \cos \psi_d - \bar{\mu}_x \sin \psi_d) \right)$$

$$\theta_d = \tan^{-1} \left(\frac{\bar{\mu}_x \cos \psi_d + \bar{\mu}_y \sin \psi_d}{\bar{\mu}_z - g} \right)$$

- It is possible to develop the same estimator starting from the coordinate-free dynamic model of a UAV and, with suitable modifications, insert the estimation within the geometric controller



- Second-order estimator

$$G_i(s) = \frac{\omega_{n,i}^2}{s^2 + 2\zeta_i\omega_{n,i}s + \omega_{n,i}^2}, i = 1, \dots, 6$$

$$\frac{d^2}{dt^2} \begin{bmatrix} \hat{f}_e(t) \\ \hat{t}_e(t) \end{bmatrix} + K_2 \frac{d}{dt} \begin{bmatrix} \hat{f}_e(t) \\ \hat{t}_e(t) \end{bmatrix} + K_2 K_1 \begin{bmatrix} \hat{f}_e(t) \\ \hat{t}_e(t) \end{bmatrix} = K_2 K_1 \begin{bmatrix} f_e(t) \\ \tau_e(t) \end{bmatrix}$$

$$\begin{bmatrix} \hat{f}_e \\ \hat{t}_e \end{bmatrix} = K_2 \left(\int_0^t - \begin{bmatrix} \hat{f}_e \\ \hat{t}_e \end{bmatrix} + K_1 \left(q - \int_0^t \begin{bmatrix} \hat{f}_e \\ \hat{t}_e \end{bmatrix} + \begin{bmatrix} mge_3 - u_T R_b e_3 \\ C^T(\eta_b, \dot{\eta}_b) \dot{\eta}_b + Q^T(\eta_b) \tau^b \end{bmatrix} dt \right) dt \right)$$

$$K_2 = 2\zeta_i\omega_{n,i}I_6$$

$$K_2 K_1 = \omega_{n,i}^2 I_6$$



- r^{th} -order estimator

$$G_i(s) = \frac{c_0}{s^r + c_{r-1}s^{r-1} + \dots + c_1s + c_0}, i = 1, \dots, 6$$

- Hurwitz polynomial

$$\sum_{i=0}^r \prod_{j=-r}^{-(i+1)} K_{-j} \frac{d^i}{dt^i} [\hat{f}_e(t)] = \prod_{i=1}^r K_i \begin{bmatrix} f_e(t) \\ \tau_e(t) \end{bmatrix}$$

$$\gamma_1(t) = K_1 \left(q - \int_0^t \begin{bmatrix} \hat{f}_e \\ \hat{\tau}_e \end{bmatrix} + \begin{bmatrix} mge_3 - u_T R_b e_3 \\ C^T(\eta_b, \dot{\eta}_b) \dot{\eta}_b + Q^T(\eta_b) \tau^b \end{bmatrix} dt \right)$$

$$\gamma_i(t) = K_i \int_0^t - \begin{bmatrix} \hat{f}_e \\ \hat{\tau}_e \end{bmatrix} + \gamma_{i-1} dt, \quad i = 2, \dots, r$$

$$\prod_{i=j+1}^r K_i = c_j, \quad j = 0, \dots, r-1$$

- The main objective of this controller is to avoid the feedback linearization of the angular part
 - At the same time, the estimation of the previously presented estimator is employed to compensate for the unknown terms (external disturbances + unmodeled dynamics)
- We consider the RPY quadrotor dynamic model

$$\begin{cases} m\ddot{p}_b = mge_3 - u_T R_b e_3 + f_e \\ M(\eta_b)\ddot{\eta}_b = -C(\eta_b, \dot{\eta}_b)\dot{\eta}_b + Q^T(\eta_b)\tau^b + \tau_e \end{cases}$$

- As usual, the planner gives the desired trajectory for the position components and the yaw angle

- Define the following reference quantities

$$\dot{\eta}_r = \dot{\eta}_{b,d} - \sigma e_\eta$$

$$\ddot{\eta}_r = \ddot{\eta}_{b,d} - \nu \dot{e}_\eta$$

$$e_\eta = \eta_b - \eta_{b,d}$$

$$\dot{e}_\eta = \dot{\eta}_b - \dot{\eta}_{b,d}$$

$$v_\eta = \dot{e}_\eta + \sigma e_\eta$$

- $\sigma > 0$ a positive coupling parameter
- The following passivity-based control can be defined for the inner (angular) loop

$$\tau^b = Q^{-T} \left(M(\eta_b) \ddot{\eta}_r + C(\eta_b, \dot{\eta}_b) \dot{\eta}_r - \hat{\tau}_e - D_o v_\eta - K_o e_\eta \right)$$
 - $D_o, K_o \in \mathbb{R}^{3 \times 3}$ positive definite gain matrices



- For the outer loop, define the following errors

$$e_p = p_b - p_{b,d}$$

$$\dot{e}_p = \dot{p}_b - \dot{p}_{b,d}$$

$$\ddot{e}_p = \ddot{p}_b - \ddot{p}_{b,d}$$

$$\mu_d = [\mu_x \quad \mu_y \quad \mu_z]^T = -\frac{1}{m}u_T R_b(\eta_{b,d})e_3 + ge_3 + \frac{1}{m}\hat{f}_e$$

- Replacing $\eta_b = e_\eta + \eta_{b,d}$ and defining $\tilde{f} = f_e - \hat{f}_e$ the linear estimation error yield

$$\ddot{p}_b = \mu_d + \frac{1}{m}u_T\delta + \frac{1}{m}\tilde{f}$$

- With δ the interconnection term as for the hierarchical controller
- As for the hierarchical controller, we forget for one moment the underactuation and we consider μ_d our input

$$\mu_d = -K_d\dot{e}_p - K_p e_p + \ddot{p}_{b,d}$$



- The closed-loop equations are

$$m\ddot{e}_p + K_d\dot{e}_p + K_p e_p = u_T\delta + \tilde{f}$$

$$M\dot{v}_\eta + (C + D_o)v_\eta + K_o e_\eta = \tilde{\tau}$$

- The right side of the linear closed-loop equation acts like an external force
 - It depends on both the quadrotor's attitude error and the estimation error
 - Boundness of the errors can be proven through Lyapunov and perturbation theories
- The right side of the angular closed-loop equation acts like an external disturbance of the mass-damping-spring system of the left side
 - This disturbance is given by the estimation error of the angular part
 - This equation can be proven to establish a **passive** relationship between $\tilde{\tau}$ and v_η
- Both closed-loop equations can be seen as mass-damping-spring systems with programmable stiffness, partially-programmable damping and given mass and inertia
 - K_p and K_d have physical meanings
 - Recalling that the linear part is the slowest, the closed-loop bandwidth for the linear part is defined
 - Upon this, the estimator's bandwidth can be tuned: it must be faster than the closed-loop linear dynamics (larger bandwidth)
 - K_o and D_o can be finally tuned
 - The coupling factor, σ , should be properly tuned since it affects the tracking results dramatically
 - The robot may vibrate for small values of σ , while larger values improve tracking
 - A common choice is $K_o = \sigma D_o$ that is equivalent to write a quadratic optimization problem



- The total thrust can be computed from the desired acceleration vector as usual

$$\bar{\mu} = [\bar{\mu}_x \quad \bar{\mu}_y \quad \bar{\mu}_z]^T = \mu_d - \frac{1}{m} \hat{f}_e$$

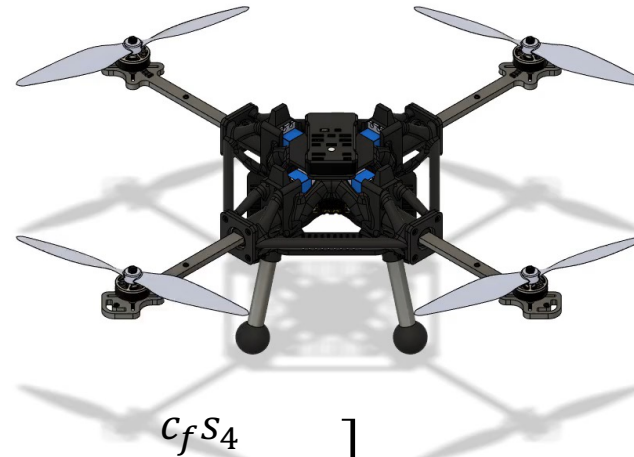
$$u_T = m \sqrt{\bar{\mu}_x^2 + \bar{\mu}_y^2 + (\bar{\mu}_z - g)^2}$$

$$\varphi_d = \sin^{-1} \left(\frac{m}{u_T} (\bar{\mu}_y \cos \psi_d - \bar{\mu}_x \sin \psi_d) \right)$$

$$\theta_d = \tan^{-1} \left(\frac{\bar{\mu}_x \cos \psi_d + \bar{\mu}_y \sin \psi_d}{\bar{\mu}_z - g} \right)$$

- Notice that the condition $\bar{\mu} = \mu_d - \frac{1}{m} \hat{f}_e = g e_3$ should be avoided as usual
- The proposed controller can be employed without the estimator of external forces following the same passages for the linear part as for the hierarchical controller

- Schematic of the considered tilting quadrotor



- Allocation matrix (**Holicopter** approach)

$$\begin{bmatrix} f_x \\ f_y \\ f_z \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -c_f s_2 & 0 & c_f s_4 \\ c_f s_1 & 0 & -c_f s_3 & 0 \\ -c_f c_1 & -c_f c_2 & -c_f c_3 & -c_f c_4 \\ 0 & -lc_f c_2 - c_m s_2 & 0 & lc_f c_4 + c_m s_4 \\ lc_f c_1 + c_m s_1 & 0 & -lc_f c_3 - c_m s_3 & 0 \\ lc_f s_1 - c_m c_1 & -lc_f s_2 + c_m c_2 & lc_f s_3 - c_m c_3 & -lc_f s_4 + c_m c_4 \end{bmatrix}}_{G_q(\alpha)} \begin{bmatrix} u_{\omega_1} \\ u_{\omega_2} \\ u_{\omega_3} \\ u_{\omega_4} \end{bmatrix} = \begin{bmatrix} G_{q,f}(\alpha) \\ G_{q,\tau}(\alpha) \end{bmatrix} u_\omega = \sum_{i=1}^4 g_{q,i}(\alpha) u_{\omega_i}$$



- Geometric dynamic model of a tilting UAV

$$\begin{cases} m\ddot{p}_b = mge_3 + R_b f^b = mge_3 + R_b G_{q,f}(\alpha) u_\omega \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b) I_b \omega_b^b + \tau^b = -S(\omega_b^b) I_b \omega_b^b + G_{q,\tau}(\alpha) u_\omega \\ \dot{\alpha} = u_\alpha \end{cases}$$

- $u_\alpha \in \mathbb{R}^4$ the control input vector for the tilting angles
- Re-arranging by collecting the inputs

$$\begin{cases} \begin{bmatrix} mI_3 & O \\ O & I_b \end{bmatrix} \begin{bmatrix} \ddot{p}_b \\ \dot{\omega}_b^b \end{bmatrix} = \begin{bmatrix} mge_3 \\ -S(\omega_b^b) I_b \omega_b^b \end{bmatrix} + \begin{bmatrix} R_b & O \\ O & I_3 \end{bmatrix} \begin{bmatrix} G_{q,f}(\alpha) & O \\ G_{q,\tau}(\alpha) & O \end{bmatrix} \begin{bmatrix} u_\omega \\ u_\alpha \end{bmatrix} = \begin{bmatrix} mge_3 \\ -S(\omega_b^b) I_b \omega_b^b \end{bmatrix} + \begin{bmatrix} R_b & O \\ O & I_3 \end{bmatrix} G_q(\alpha) u_\omega \\ \dot{R}_b = R_b S(\omega_b^b) \\ \dot{\alpha} = u_\alpha \end{cases}$$

- The allocation matrix has rank less than 6, therefore the system appears to be underactuated
 - This is because u_α acts at a higher differential level compared to u_ω ; therefore, a direct inversion at an acceleration level involves the inputs u_ω only, resulting in loss of controllability for the system



- A possible way to circumvent these difficulties is to resort to a dynamic output linearization scheme and seek to invert the input–output map at a higher differential level where inputs u_α will appear explicitly
- Notice that

$$\frac{dG_q(\alpha)u_\omega}{dt} = G_q(\alpha)\dot{u}_\omega + \sum_{i=1}^4 \frac{\partial g_{q,i}(\alpha)}{\partial \alpha} u_{\omega_i} \dot{\alpha} = G_q(\alpha)\dot{u}_\omega + \sum_{i=1}^4 \frac{\partial g_{q,i}(\alpha)}{\partial \alpha} u_{\omega_i} u_\alpha$$

- Writing the geometric dynamic model of a tilting UAV at jerk level yields

$$\left\{ \begin{aligned} \begin{bmatrix} mI_3 & O \\ O & I_b \end{bmatrix} \begin{bmatrix} \ddot{p}_b \\ \ddot{\omega}_b^b \end{bmatrix} &= \begin{bmatrix} 0 \\ -S(\dot{\omega}_b^b)I_b\omega_b^b - S(\omega_b^b)I_b\dot{\omega}_b^b \end{bmatrix} + \begin{bmatrix} \dot{R}_b & O \\ O & O \end{bmatrix} G_q(\alpha)u_\omega + \begin{bmatrix} R_b & O \\ O & I_3 \end{bmatrix} \left(G_q(\alpha)\dot{u}_\omega + \sum_{i=1}^4 \frac{\partial g_{q,i}(\alpha)}{\partial \alpha} u_{\omega_i} u_\alpha \right) \\ \dot{R}_b &= R_b S(\omega_b^b) \\ \dot{\alpha} &= u_\alpha \end{aligned} \right.$$

- Rearranging yields

$$\left\{ \begin{aligned} \begin{bmatrix} \ddot{p}_b \\ \ddot{\omega}_b^b \end{bmatrix} &= \begin{bmatrix} mI_3 & O \\ O & I_b \end{bmatrix}^{-1} \left(\begin{bmatrix} 0 \\ -S(\dot{\omega}_b^b)I_b\omega_b^b - S(\omega_b^b)I_b\dot{\omega}_b^b \end{bmatrix} + \begin{bmatrix} R_b S(\omega_b^b) & O \\ O & O \end{bmatrix} G_q(\alpha)u_\omega \right) + \begin{bmatrix} mI_3 & O \\ O & I_b \end{bmatrix}^{-1} \begin{bmatrix} R_b & O \\ O & I_3 \end{bmatrix} \left[G_q(\alpha) \sum_{i=1}^4 \frac{\partial g_{q,i}(\alpha)}{\partial \alpha} u_{\omega_i} \right] \begin{bmatrix} \dot{u}_\omega \\ u_\alpha \end{bmatrix} \\ \dot{R}_b &= R_b S(\omega_b^b) \\ \dot{\alpha} &= u_\alpha \end{aligned} \right.$$



- This can be re-written a

$$\left\{ \begin{array}{l} \begin{bmatrix} \ddot{p}_b \\ \ddot{\omega}_b^b \end{bmatrix} = b(R_b, \omega_b^b, \dot{\omega}_b^b, \alpha, u_\omega) + A(R_b, \alpha, u_\omega) \begin{bmatrix} u_v \\ u_\alpha \end{bmatrix} \\ \dot{R}_b = R_b S(\omega_b^b) \\ \dot{\alpha} = u_\alpha \\ \dot{u}_\omega = u_v \end{array} \right.$$

- The new input $\dot{u}_\omega$ is the dynamic extension of the former (and actual) input u_ω
- The new input matrix $A(R_b, \alpha, u_\omega) \in \mathbb{R}^{6 \times 8}$, where now the second column-block is not made by zero matrices any more
 - This matrix has rank 6 as long as $u_\omega \neq 0$; in other words, full rankness of $A(R_b, \alpha, u_\omega)$ is ensured by preventing the propellers to stop their spinning motion, that is reasonable
- When $A(R_b, \alpha, u_\omega)$ is full rank, we can exploit feedback linearization to design the controller
 - If the task is to track the desired position, velocity, and acceleration for the translational and angular part, a classic PID controller plus jerk feed-forward can be employed
 - Since the system is redundant with respect to the given task, it is possible to employ the redundancy to optimize additional criteria (e.g., minimizing the norm of u_ω for energy consumption)



- Another approach (Voliro approach)

$$\begin{cases} m\ddot{p}_b = mge_3 + R_b f^b \\ \dot{R}_b = R_b S(\omega_b^b) \\ I_b \dot{\omega}_b^b = -S(\omega_b^b) I_b \omega_b^b + \tau^b \end{cases}$$

- We can directly design the control through f^b and τ^b (e.g., by using feedback linearization)
- We can compute the inputs u_ω and α by transforming the nonlinear allocation problem into a linear problem through a variable transformation

$$\begin{aligned} u_{v,i} &= c_f u_{\omega_i} c_i \\ u_{l,i} &= c_f u_{\omega_i} s_i, \quad i = 1, \dots, 4 \end{aligned}$$

- This yields re-writing the allocation matrix as

$$\begin{bmatrix} f_x \\ f_y \\ f_z \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & -1 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & -l & -\frac{c_m}{c_f} & 0 & 0 & l & \frac{c_m}{c_f} \\ l & \frac{c_m}{c_f} & 0 & 0 & -l & -\frac{c_m}{c_f} & 0 & 0 \\ -\frac{c_m}{c_f} & l & \frac{c_m}{c_f} & -l & -\frac{c_m}{c_f} & l & \frac{c_m}{c_f} & -l \end{bmatrix} \begin{bmatrix} u_{v,1} \\ u_{l,1} \\ u_{v,2} \\ u_{l,2} \\ u_{v,3} \\ u_{l,3} \\ u_{v,4} \\ u_{l,4} \end{bmatrix} = G_{q,static} u(\alpha, u_\omega)$$

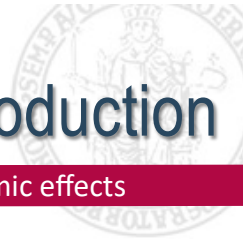
- It is possible to retrieve $u(\alpha, u_\omega)$ from f^b and τ^b by pseudo-inverting $G_{q,static}$
 - It is possible to exploit the null space of the pseudoinverse by optimizing further criteria
- Obtaining the actual rotor speed and the rotor orientation is straightforward

$$u_{\omega_i} = \frac{1}{c_f} \sqrt{u_{l,i}^2 + u_{v,i}^2}, i = 1, \dots, 4$$
$$\alpha_i = \text{atan2}(u_{l,i}, u_{v,i})$$

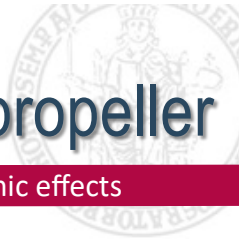
- Drawbacks
 - Allocating the actuator commands using the pseudoinverse may result in inadmissible rotor speeds that are higher than physically possible
 - This allocation does not account for the slower dynamics of the tilting angles
 - This leads to forces and moments being produced in directions that have not been commanded



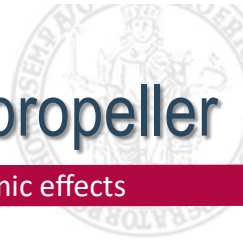
- Further approaches
 - Nonlinear least-squares optimization
 - The computation time required and the available computation resources on board the platform may inhibit such a solution
 - Nonlinear MPC attitude controller to generate near-optimal rotor-speed and tilting-angle command
 - Computationally expensive
 - ...



- UAVs and UAMs are usually involved in applications that require flying close to different structures, objects, and obstacles
- These devices can also grasp objects or make contacts with the environment for inspection tasks
- In all cases, the airflow produced by the rotors is constrained by the environment
 - This results in a change of the total thrust and torque



- This is the effects that a UAV is subject to when it flies closes to the ground
- The most employed theoretical result for the ground effect is based on the so-called **potential aerodynamic assumption**
 - This assumption considers the fluid
 - Inviscid
 - Incompressible
 - Irrotational
 - Steady
- In order to model the effect of a surface like the ground, the **method of images** is employed
 - This method consists in placing a virtual rotor on the opposite side of the surface and at the same distance
 - If one rotor is placed at $[0 \quad 0 \quad h]^T$, the other rotor is placed at $[0 \quad 0 \quad -h]^T$



- Let v_{IGE} and T_{IGE} be the induced velocity and the thrust of the rotor inside the ground effect, respectively
- Let v_{OGE} and T_{OGE} be the induced velocity and the thrust of the rotor outside the ground effect, respectively

- From power preserving theory, the following relation holds

$$v_{IGE} T_{IGE} = v_{OGE} T_{OGE}$$

- From the previous one and from aerodynamic considerations

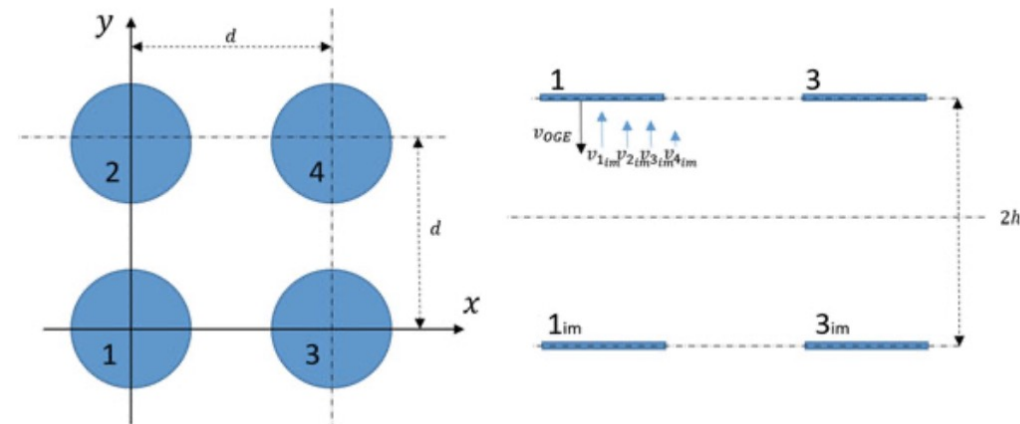
$$\frac{T_{IGE}}{T_{OGE}} = \frac{1}{1 - \left(\frac{\rho}{4z}\right)^2}$$

- With ρ the radius of the propeller, and z the height of the propeller from the ground
- Notice that for $\frac{z}{\rho} = 2$, the ratio is 1.016
 - Therefore, this formula predicts that the ground effect is negligible when the rotor is more than one diameter off the ground

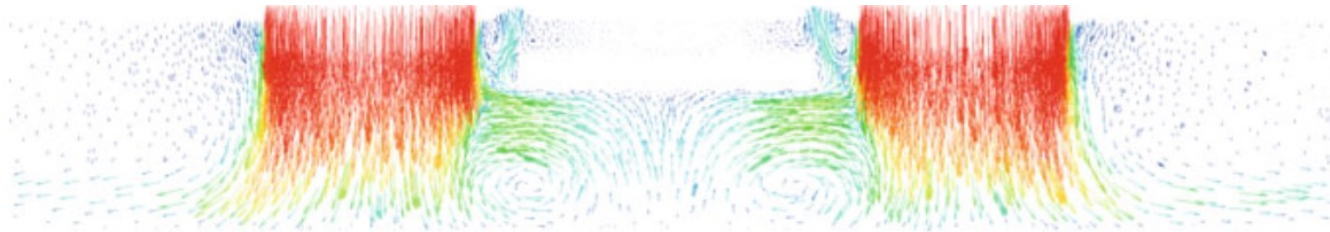
$$\frac{z}{\rho} > 2 \Rightarrow z > 2\rho$$

- In the quadrotor case, each propeller has an image source of equal strength placed below the correspondent rotor at a distance equal to twice the height of the rotor
- It is necessary to re-write the expression of the velocity induced inside the ground effect to take into account the effects of the other rotors
- Supposing that the power remains constant for the complete multirotor, the following expression holds

$$\frac{u_{T,IGE}}{u_{T,OGE}} = \frac{1}{1 - \left(\frac{\rho}{4z}\right)^2 - \frac{z\rho^2}{\sqrt{(d^2 + 4z^2)^3}} - \frac{z\rho^2}{2\sqrt{(2d^2 + 4z^2)^3}}}$$

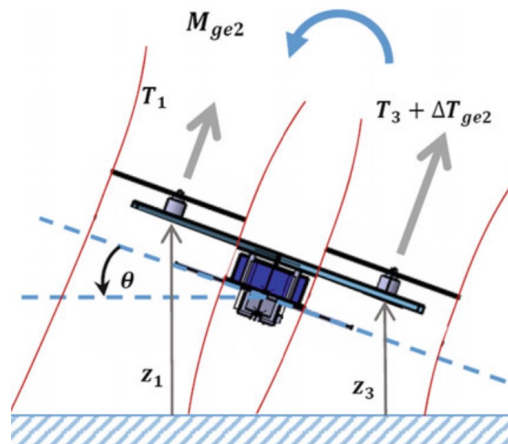


- This picture shows a **computational fluid dynamics** (CFD) simulation of two rotors close to the ground



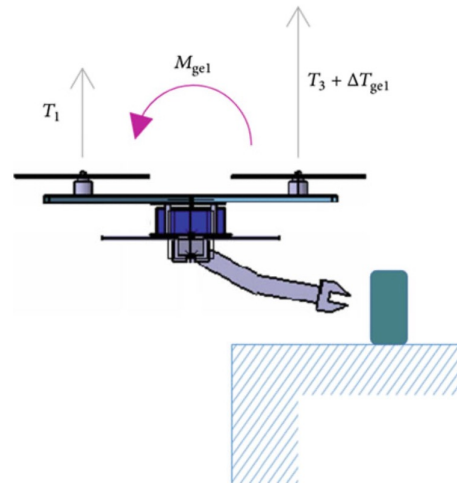
- The simulation shows that the airflow in the middle of the space between the two rotors is reverted after hitting the ground
- The airflow hits the multicopter frame producing an extra lifting force

- This picture shows the hovering multirotor subject to an attitude perturbation



- Since the increment in the rotor thrust depends on the distance of each rotor to the ground, and since the distances above are different, a disturbance moment, M_{ge} , will appear
 - In the condition above, this disturbance indeed tries to stabilize the vehicle

- This picture shows a UAM hovering or it is flying at low speed but only a part of the rotors are subject to the ground effect



- This asymmetrical problem produces a different aerodynamic effect in each rotor
- In the figure, the moment M_{ge} will arise and will try to rotate the multirotor, preventing the grasp of the object

- This effect arises when a multirotor needs to approach some objects from below (i.e., bridges, tunnels, ...)
- In the single rotor case, it happens that the thrust increases significantly closes to the ceil, pushing the rotor even closer to it
 - Therefore, it is important to take into account such an effect to avoid crashes
 - The increased thrust is a consequence of the increase of the propeller rotational velocity, caused by the **vacuum effect** which decreases the propeller drag when it is very close to the ceil, so that the propeller rotates faster

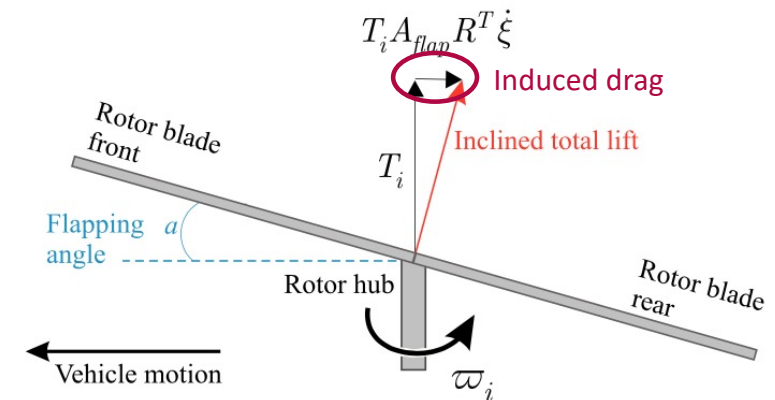
- The effect can be approximated by the following formula

$$\frac{T_{ICE}}{T_{OCE}} = \frac{1}{1 - \frac{1}{k_1} \left(\frac{\rho}{z + k_2} \right)^2}$$

- T_{ICE} and T_{OCE} are the single thrust inside and outside the ceiling effect, respectively
 - k_1, k_2 are parameters to tune
- Therefore, through this effect, it is possible to develop more thrust for the same spent power
 - This results in less energy consumption and more flight time (if crashes are avoided ...)
 - This energy saving result is particularly interesting in those applications requiring to maintain a contact with the ceiling

- Wall effect
 - This effect appears when the UAV is flying close to a wall
 - In case of flat UAVs (fixed propellers), the wall effect is negligible since the propellers and the wall are orthogonal
 - This may be not true in case of tilted or tilting UAVs
- Pipe effect
 - In oil and gas refineries, the environments cluttered and full of pipelines
 - Flying over a pipe, or close to a pipe, or approaching a pipe from below have the same effects of the ground, wall, and ceil effects, respectively
 - However, the size and the shape of the pipes may mitigate a lot these effects

- During the flight, the high velocities of the propellers causes the flapping of the blades
 - Flapping means an up-and-down oscillation due to some elasticity components of the blade
- The flapping causes a different drag force that is often omitted or neglected
 - The reason is that these forces are dissipative and contribute to the stability of the vehicle during hovering
- Besides, since these drag forces are equally distributed in hovering, they can be responsible for the aerodynamic torque causing the yaw
- However, during a trajectory, these forces may become a disturbance
- Aerodynamic drag due to blade flapping affects the UAV during the forward flight
 - In detail, the advancing rotor blade has a higher velocity than the rear one and it will generate a lift force
 - In turn, this creates a bigger induced drag than the rear blade
 - This results in a net force that is opposite to the current forward motion



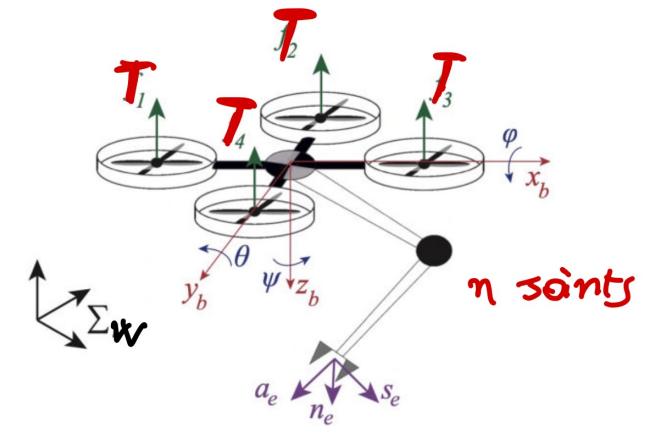
- Let be the following vector the generalised joint vector

$$\zeta = \begin{bmatrix} p_b \\ \eta_b \\ q_1 \\ \vdots \\ q_n \end{bmatrix} \in \mathbb{R}^{6+n}$$

- Let consider the kinematics of the robot with respect to the body frame of the UAV and then expressed in the world fixed frame

$$T_e = T_b T_e^b$$

- $T_e \in SE(3) = \mathbb{R}^3 \times SO(3)$ the end-effector frame in the world frame
- $T_b = \begin{bmatrix} R_b & p_b \\ 0^T & 1 \end{bmatrix} \in SE(3)$ the UAV body frame in the world frame
- $T_e^b \in SE(3)$ the end-effector frame in the UAV body frame



- The differential kinematics is

$$\begin{bmatrix} \dot{p}_b \\ \omega_b^b \\ \dot{p}_e^b \\ \omega_e^b \end{bmatrix} = \begin{bmatrix} I_3 & O_3 & O \\ O_3 & Q(\eta_b) & O \\ O & O & J(q) \end{bmatrix} \dot{\zeta}$$

- $\dot{p}_e^b \in \mathbb{R}^3$ the linear velocity of the end-effector in the body frame
 - $\omega_e^b \in \mathbb{R}^3$ the angular velocity of the end-effector in the body frame
- Now, put a frame at each link's centre of mass
 - Let $p_{l_i} \in \mathbb{R}^3$ be the position of the center of mass of the i -th link in the world fixed frame

$$p_{l_i} = p_b + R_b p_{bl_i}^b, \quad i = 1, \dots, n$$

- $p_{bl_i}^b \in \mathbb{R}^3$ is the position of the center of mass of the link in the body frame

- The following expressions hold for $i = 1, \dots, n$

$$\dot{p}_{bl_i} = J_p^{(l_i)} \dot{q}$$

$$\omega_{bl_i} = J_o^{(l_i)} \dot{q}$$

- Therefore, deriving p_{l_i} and ω_{l_i} with respect to time yields

$$\dot{p}_{l_i} = \dot{p}_b - S(R_b p_{bl_i}) \omega_b + R_b J_p^{(l_i)} \dot{q}$$

$$\omega_{l_i} = \omega_b - R_b J_o^{(l_i)} \dot{q}$$

- The dynamic model can be derived through the Euler-Lagrangian equations, considering the generic element of the generalised joint vector, ζ , and the associated generalised force, u_i
- The Lagrangian is $\mathcal{L} = K - U$
- The kinetic energy is $K = K_b + \sum_{i=1}^n K_{l_i}$
 - $K_b = \frac{1}{2} m \dot{p}_b^T p_b + \frac{1}{2} \dot{\eta}_b^T Q^T I_b Q \dot{\eta}_b$ is the UAV kinetic energy
 - $K_{l_i} = \frac{1}{2} m_{l_i} \dot{p}_{l_i}^T p_{l_i} + \frac{1}{2} \omega_{l_i}^T R_b R_{l_i}^b H_{l_i} R_b^{l_i} R_b^T \omega_{l_i}$ is the kinetic energy of the i -th link
 - m_{l_i} mass of the i -th link
 - $R_{l_i}^b \in SO(3)$ is the rotation matrix between the centre of mass of the i -th link and the body frame
 - The total kinetic energy is given by the mass matrix whose components are

$$\begin{aligned}
 B_{11} &= \left(m_b + \sum_{i=1}^n m_{l_i} \right) I_3 \\
 B_{22} &= Q^T H_b Q + \sum_{i=1}^n \left(m_{l_i} T_b^T S(R_b p_{bl_i}^b)^T S(R_b p_{bl_i}^b) T_b \right. \\
 &\quad \left. + Q^T R_{l_i}^b H_{l_i} R_b^{l_i} Q \right) \\
 B_{33} &= \sum_{i=1}^n \left(m_{l_i} J_P^{(l_i)T} J_P^{(l_i)} + J_O^{(l_i)T} R_{l_i}^b H_{l_i} R_b^{l_i} J_O^{(l_i)} \right) \\
 B_{12} &= B_{21}^T = - \sum_{i=1}^n \left(m_{l_i} S(R_b p_{bl_i}^b) T_b \right) \\
 B_{13} &= B_{31}^T = \sum_{i=1}^n \left(m_{l_i} R_b J_P^{(l_i)} \right) \\
 B_{23} &= B_{32}^T = \sum_{i=1}^n \left(Q^T R_{l_i}^b H_{l_i} R_b^{l_i} J_O^{(l_i)} \right. \\
 &\quad \left. - m_{l_i} T_b^T S(R_b p_{bl_i}^b)^T R_b J_P^{(l_i)} \right),
 \end{aligned}$$

- The total potential energy is $U = U_b + \sum_{i=1}^n U_{l_i}$
 - $U_b = mge_3^T p_b$ is the UAV potential energy
 - $U_{l_i} = m_{l_i}ge_3^T (p_b + R_b p_{bl_i}^b)$ is the link potential energy
- Computing the Christoffel symbols as usual, the dynamic model of a UAM in compact form is

$$B(\zeta)\ddot{\zeta} + C(\zeta, \dot{\zeta})\dot{\zeta} + \bar{g}(\zeta) = u$$

- $\bar{g}(\zeta) = \frac{\partial U(\zeta)}{\partial \zeta}$ is the gravity vector
 - $u \in \mathbb{R}^{n+6}$ is not the real input vector which is computed below
- Let define the real input vector as

$$f = \begin{bmatrix} u_T \\ \tau^b \\ \tau_q \end{bmatrix} \in \mathbb{R}^{4+n}$$

- This in case of a flat UAV configuration and a fully-actuated manipulator
 - $\tau_q \in \mathbb{R}^n$ are the manipulator torques

- The relationship between u and f is given by

$$u = \Omega f$$

- $\Omega = \text{diag}\{R_b e_3, Q^T, I_n\} \in \mathbb{R}^{(n+6) \times (n+4)}$
- $B(\zeta)\ddot{\zeta} + C(\zeta, \dot{\zeta})\dot{\zeta} + \bar{g}(\zeta) = \Omega f$

- The UAM is seen as a unique entity
- It is a classical mechanical system with dynamic model $B(\zeta)\ddot{\zeta} + C(\zeta, \dot{\zeta})\dot{\zeta} + \bar{g}(\zeta) = \Omega f$
- If the UAV is tilted or tilting, the mapping given by Ω may be bijective (fully actuation)
 - Otherwise, as in the case of quadrotor, we have that $rank(\Omega) < \dim[\zeta]$ and the system is underactuated
 - In this case, one solution is to design the controller for the entire control vector, u
 - Then, the pseudo-inverse of Ω can be calculated since $\Omega^T \Omega$ is always invertible
 - Otherwise, a hierarchical controller for the UAM may be devised

- The UAV and the manipulator are seen as two separated entities
 - The flight is decoupled from the manipulation
- The UAV is controlled independently from the manipulator
 - Each device has its own controller implementing suitable estimators to compensate for reciprocal interactions and disturbances